Digital Infrastructure Construction and Industrial Chain Resilience: Empirical Evidence from High-tech industry in China

Xia Zhou[1], Shen'ao Liu [1*]

[1]School of Urban Economics and Management, Beijing University of Civil Engineering and Architecture, Beijing, China

*** Correspondence:**
Shen'ao Liu
liusa7255@163.com

Keywords: Digital Infrastructure; Industrial Chain Resilience; High-tech Industrial Chain Resilience; High-Tech Industry; Resilience

Abstract

Currently, China's high-tech industrial chain faces a "dual challenge" both internally and externally. Enhancing high-tech industrial chain resilience has become a critical issue for safeguarding China's industrial security. As a crucial foundation for the digital era, digital infrastructure construction provides significant support for improving high-tech industrial chain resilience. Based on panel data from 31 provinces, municipalities, and autonomous regions in China from 2008 to 2021, this study employs panel regression model and mediation effect model to examine the mechanisms through which digital infrastructure construction influences high-tech industrial chain resilience. The empirical results indicate that digital infrastructure construction has a significant positive impact on high-tech industrial chain resilience, with an average effect of 7.2%. This impact is mediated through three pathways: industrial co-agglomeration, innovation efficiency, and factor allocation level. Overall, future efforts should focus on optimizing the layout planning and policy system of digital infrastructure construction, strengthening the collaborative cooperation of related industries, and continuously accelerating the digital transformation of the whole chain enterprises. This will help build a high-quality ecosystem for high-tech industrial chain development, fostering more cohesive, dynamic, resilient, and competitive high-tech industrial chain.

1 Introduction

The ongoing wave of technological revolution and industrial transformation is profoundly reshaping modern industrial development, which increasingly emphasizes regional division of labor, trade liberalization, and scientific and technological support. This has ushered in the era of "chain clusters". However, in recent years, as both internal and external situations have become increasingly severe, China's industrial chain development faces a "dual challenge". Externally, factors such as the Trade disputes between China and the United States, the COVID-19 pandemic, and geopolitical conflicts have introduced numerous global uncertainties that threaten industrial chain security (Du et al., 2023). The global manufacturing sector is increasingly shifting toward localization and shorter supply chains. According to the UNCTAD (UN Trade and Development) Cross-Border M&A (Mergers & Acquisitions) Database, global manufacturing multinational corporations have become more cautious in their investments, with the scale of mergers and acquisitions dropping from 394.208 billion dollars in 2015 to 142.313 billion dollars in 2022, a decline of 63.9% (Figure 1). Internally, while China's

This is a provisional file, not the final typeset article

industrial chains occupy a high position in the global value chain in terms of "quantity", they still lag behind developed countries in terms of "quality", with key core technologies facing "bottleneck" issues. According to a 2018 Ministry of Industry and Information Technology (MIIT) study of more than 130 key basic materials from more than 30 large enterprises in China, the results show that 32% of key materials are still unavailable domestically, and 52% rely on imports. Additionally, over 95% of high-end chips for computers and servers and more than 70% of processors for smart devices are imported. In October 2022, the 20th National Congress of the Communist Party of China emphasized the need to "enhance the resilience and security of industrial and supply chains", making the strengthening of industrial chain resilience a key issue for China's high-quality economic development.

On the other hand, the widespread application of new-generation digital technologies such as artificial intelligence, big data, the Internet of Things (loT), and 5G communication has driven the rapid development of the digital economy. According to the China Digital Economy Development Research Report (2024) released by the China Academy of Information and Communications Technology in April 2024, China's scale of digital economy development reached 53.9 trillion yuan in 2023, accounting for 42.8% of GDP (Gross Domestic Product), far exceeding China's GDP growth rate (Figure 2). As a critical foundation for the digital economy development, digital infrastructure construction provides essential support for the digital economy, playing a pivotal role in enabling the integration of the digital economy with the real economy. According to data from China Academy of Information and Communications Technology, the length of optical cable lines in China has increased to 6,432 kilometers, a 3.68-fold increase from 2013. The number of 5G base stations has reached 3.377 million, and the national computing power hub nodes under the "East Data, West Computing" project are steadily advancing, with the core computing power industry exceeding 1.8 trillion yuan. As digital infrastructure construction continues to expand and improve, its connection with socio-economic development has become increasingly close, gradually integrating into various segments of the industrial chain and value chain. This has provided significant "digital dividends" for China's technological system innovation and enterprise reform, while also permeating government, public, and social governance reforms (Liu et al., 2020). It has effectively unleashed economic growth potential and facilitated the transition from old to new growth drivers.

High-tech industries, characterized by high added value, high returns, low energy consumption, and low pollution, have always been catalysts for rapid regional economic development. They are also powerful pillars for improving economic structure, enhancing national technological development levels, and boosting comprehensive competitiveness (Ren & Tang, 2024). According to the China High-tech Industry Yearbook and the World Bank database, it shows that as of 2022, the number of enterprises in China's high-tech industry exceeded 50,000, the value added of high-tech industry above designated size increased by 7.4% compared with the previous year, and the actual use of foreign capital was 444.9 billion yuan with an overall positive development trend. However, compared with the developed countries, there is still a big gap that some of the core indicators are ranked in the lower middle class. The pace of development of high-tech industry is still relatively slow and characterized by the predicament of "low-end locking" (Figure 3). Against the backdrop of the increasingly prominent strategic position of digital infrastructure construction, how to leverage its power to build more cohesive, dynamic, resilient, and competitive high-tech industrial chain has become a focal issue for ensuring the security and stability of China's industrial and supply chains. Therefore, this paper focuses on the two core subjects of digital infrastructure construction and high-tech industrial chain resilience, aiming to reveal the underlying mechanisms and propose practical policy recommendations for enhancing the resilience of China's high-tech industrial chain.

This is a provisional file, not the final typeset article

The remainder of the paper is organized as follows: Section 2 provides a literature review. Section 3 presents the research hypotheses. Section 4 discusses the research methodology and data sources. Section 5 presents the empirical results, including benchmark regression, instrumental variable tests, robustness checks, mediation effect tests, and spatial spillover effect tests. Section 6 concludes with policy implications. The potential contributions of this study are as follows: (1) In terms of research perspective, this paper shifts the focus from "industry" to "industrial chain", enriching the theoretical and empirical research on industrial chain resilience from both macro and micro perspectives, providing a foundation for future research. (2) In terms of research content, this paper innovatively combines digital infrastructure construction with industrial chain resilience, focusing on the high-growth high-tech industry. This broadens new pathways for incremental development and contributes to the goal of high-quality economic development in China.

2 Literature Review

2.1 Research on the Economic Effects of Digital Infrastructure Construction

The academic understanding of digital infrastructure construction is primarily based on the definition of "new infrastructure" by National Development and Reform Commission: new infrastructure is a system that, guided by new development concepts, driven by technological innovation, and based on information networks, meets the needs of high-quality development by providing digital transformation, intelligent upgrading, and integrated innovation services. The main components include information infrastructure, integrated infrastructure, and innovation infrastructure. Digital infrastructure breaks traditional geographical barriers, accelerates the advancement of information technology and the rapid flow of knowledge across regions, and provides a continuous source of momentum for China's economic development. Research on its economic effects has gradually become a hot topic in academia. From a macro perspective, digital infrastructure construction has a significant positive impact on regional coordinated development (Li et al., 2024), industrial structure upgrading (Wu et al., 2023), foreign trade exports (Zhou et al., 2022), labor employment rates (Zhou et al., 2024), and factor allocation efficiency (Pan & Yang, 2024). From a micro perspective, digital infrastructure construction positively influences innovation capabilities (Tian & Lu, 2023), entrepreneurial activity (Schade & Schuhmacher, 2022), enterprise performance growth (Queiroz et al., 2020), and labor income (Whitacre et al., 2014). Many scholars have used mediation effect models and difference-in-differences models to verify these effects. On the other hand, the rapid construction of digital infrastructure can also lead to digital divides and socio-spatial inequalities. For example, Zhang & Sun (2023) focused on China's administrative border regions and found that digital infrastructure construction can promote economic convergence between border regions and central regions, alleviating economic disparities. Nie & Wan (2023) found that information infrastructure investment significantly widens the urban-rural income gap and deepens social inequality. Jiang & Jin (2024) argued that the impact of digital infrastructure construction on the urban-rural income gap exhibits a threshold effect.

2.2 Research on the Concept, Measurement, and Influencing Factors of Industrial Chain Resilience

The concept of industrial chain resilience is largely derived from economic resilience. Since Holling (1973) first introduced the term "resilience" into the field of ecology, defining it as the ability of a system to recover and reorganize in the face of change, it has laid the foundation for resilience research. Later, under the influence of Walker's (2004) evolutionary resilience theory, many scholars extended the concept of "resilience" to social-ecological systems (Folke, 2006) and gradually expanded it to the field of spatial economics. Simmie & Martin (2010) summarized the relevant

research and innovatively proposed a regional economic resilience evolution framework of "reorganization-development-maintenance-release", emphasizing that economic resilience is a dynamic equilibrium of adaptation and adjustment capabilities, mainly including resistance, recovery, adaptability, and renewal (Martin & Sunley, 2015). This definition has been widely endorsed by scholars and serves as an important basis for understanding the industrial chain resilience in this paper.

The measurement of industrial chain resilience mostly draws on methods for measuring economic resilience, which can be broadly divided into four categories: First, core variable measurement methods. Chen & Wen (2023) used network topology core indicators to express the comprehensive ability of industrial chains to resist and recover from risks. Second, comprehensive index system measurement methods. Wu et al. (2023) used the 2F-PT-TOPSIS model to select 13 indicators to measure the resilience of coal industrial chain and supply chain in Shaanxi Province, China. Le (2023) updated this approach by selecting 23 tertiary indicators from four aspects: risk resistance, organizational adaptability, autonomous control, and leading competitiveness to measure the resilience of provincial industrial chain in China. Third, input-output measurement methods based on production networks. Li et al. (2022) used China's input-output tables and the PR algorithm to calculate the global industrial chain embedding probability under industrial chain division scenarios to reflect the industrial chain resilience. Ma et al. (2023) used the Eora MRIO multi-regional input-output table and combined the PR algorithm with a social network model to construct a global index of industrial chain resilience for various countries. Fourth, complex network attribute measurement methods. Crespo et al. (2014) focused on regional industrial knowledge networks and used degree distribution and degree correlation as important network attributes to reflect the industrial chain resilience.

The influencing factors of industrial chain resilience mainly focus on digital transformation, internal industrial structure, and technological innovation capabilities. For example, Huang et al. (2023) first used the fsQCA method to analyze the complex causal relationship between new industrialization and the resilience of the manufacturing industrial chain from a macro perspective, and then used the DEMATEL-ISM-BN model to explain the internal mechanism from a micro perspective. Yang & Liu (2024) used a mediation effect model to find that digital infrastructure construction can enhance the industrial chain resilience by optimizing industrial structure, improving capacity utilization, promoting industrial co-agglomeration, and enhancing regional innovation efficiency. Meng (2023) used a multi-period difference-in-differences method to empirically test that the "chain leader system" significantly promotes the industrial chain resilience by leveraging mechanisms such as responsibility allocation, mobilization, and factor guarantee.

2.3 Research on the Impact of Digital Infrastructure Construction on High-Tech Industries

Existing research on digital infrastructure construction and high-tech industries primarily focuses on the digital economy as the explanatory variable, examining its impact on the innovation capabilities, innovation performance, and innovation efficiency of high-tech industries. For example, Tan (2024) used a three-stage DEA model to find that the innovation efficiency of high-tech industries is positively correlated with digital economy development: for every 1% increase in the digital economy development index, innovation efficiency increases by 0.2102%. Wan et al. (2023) found that digital technology significantly promotes innovation in China's high-tech industries, but the mediation effects of R&D resource investment and technology diffusion exhibit threshold effects. Liu et al. (2024) constructed a spatial Durbin model and a semi-parametric spatial lag model for empirical analysis, finding that the partial derivative of the digital economy's impact on high-tech

This is a provisional file, not the final typeset article

industry innovation shows early fluctuations and later stability. Zhang et al. (2024) used the Hicks-Moorsteen index method to examine the impact of the digital economy on the total factor productivity and its decomposition efficiency in China's high-tech industries. The threshold effect shows that the innovation efficiency of high-tech industries reaches its optimal level when the digital economy composite index is between 2.611 and 2.747.

3 Research Hypothesis

Digital infrastructure construction, as a new type of infrastructure that integrates traditional infrastructure with modern information and digital technologies, enables the integration of data elements with traditional production factors. Through the penetration and integration of digital and real economies and network interconnection, it strengthens the development capabilities of various nodes in the high-tech industrial chain and the connections between them, thereby enhancing high-tech industrial chain resilience (Figure 4). Specifically, first, the rapid construction of information infrastructure such as broadband networks, 5G, and the Internet eliminates geographical barriers, reduces information acquisition and transaction costs, simplifies redundant information links, and improves the efficiency of various economic activities. This breaks the "information silo" dilemma between upstream and downstream segments of the high-tech industrial chain (Liao et al., 2016), enabling cross-temporal communication between different enterprise entities on the chain. It ensures the smooth operation of different production and circulation segments of the high-tech industrial chain, significantly improving the development efficiency and flexibility of the industrial chain. This allows the high-tech industrial chain to respond quickly to internal and external risks and shocks, effectively dispersing shocks and mitigating economic fluctuations, thereby enhancing high-tech industrial chain resilience. Second, according to Metcalfe's Law, digital infrastructure construction has network externalities, enabling the integration of digital technologies, data elements, and capital inputs to achieve real-time connectivity and effective integration (Ma et al., 2023). This accelerates the digital transformation of the high-tech industrial chain. During this process, decision flows, human flows, logistics, capital flows, and technology flows within the industrial chain gradually achieve digital transmission, accelerating the digital reshaping of various business processes (Ahmad et al., 2021). This ultimately drives the comprehensive digital transformation of the high-tech industrial chain toward a complex, multi-dimensional spatial network structure. This effectively enhances the ability of high-tech industrial chain to disperse risk shocks, significantly weakening the transmission of risk shocks along the industrial chain, thereby increasing the stability of the high-tech industrial chain. Third, empowered by digital infrastructure construction, the ability of high-tech industrial chain to collect and interpret real-time data is significantly improved. Digital infrastructure construction helps the high-tech industrial chain obtain real-time data on production, processing, circulation, consumption, and distribution, such as production and processing data, enterprise operation data, user data, and payment data. It conducts vertical penetration and in-depth analysis of this information, solving the "black box" problem of supply and demand information. Therefore, the high-tech industrial chain can autonomously and dynamically adjust its production plans to improve management effectiveness and develop "localized" business models based on user needs (Kangmin et al., 2023), expanding the innovation space of the high-tech industrial chain and thereby enhancing its resilience. Based on the above analysis, this paper proposes Hypothesis 1:

H1: Digital infrastructure construction has a positive impact on high-tech industrial chain resilience.

Digital infrastructure construction expands the advantages of regional digital technology development, effectively reducing the production costs of high-tech industrial chain and the communication and coordination costs between upstream and downstream enterprises, creating

economies of scale. According to the center-periphery theory, various production factors from outside the region will quickly flow into regions with technological advantages and economies of scale to access abundant development opportunities and economic resources, continuously promoting the dynamic evolution of spatial agglomeration and diffusion (Zeng et al., 2024). This fosters the continuous agglomeration of high-tech industrial enterprises. On the other hand, digital infrastructure construction investments are more closely related to R&D technologies such as 5G, big data, artificial intelligence, and blockchain, which will generate demand for necessary productive services such as technology R&D, network operations, hardware and software design, industrial data collection-storage-analysis-modeling, smart logistics, and platform operations. This creates more employment opportunities in related productive service industries, covering real estate, finance, transportation, warehousing, postal services, scientific research, technical services, geological exploration, information transmission, computer services, and software (Guo et al., 2020). Thus it promotes the agglomeration of productive service industries. Digital infrastructure construction serves as a carrier for the agglomeration and circulation of "data" elements, connecting high-tech industries and productive service industries through digital information-sharing platforms, promoting the co-agglomeration of the two industries. From a production perspective, industrial co-agglomeration provides a favorable external environment for cross-industry knowledge dissemination, improving knowledge transfer efficiency, reducing knowledge dissemination costs, and strengthening knowledge spillover effects (Grieser et al., 2022). It enhances the absorptive capacity of the high-tech chain to external economic fluctuations. Productive service industries can provide more professional and refined services to high-tech industries, promoting the emergence of more new technologies, products, and development models (Chen, 2019), thereby enhancing the path-recreation capability of high-tech industrial chain. From a circulation and consumption perspective, the collaborative exchange between productive service industries and high-tech industries can generate "mutual assistance" effects, enhancing market accessibility and enabling the production of specialized and personalized products based on consumer demand. This provides a continuous source of innovation for high-tech industrial chain (Orenstein, 2021). Based on the above analysis, this paper proposes Hypothesis 2:

H2: Digital infrastructure construction fosters industrial co-agglomeration, thereby enhancing high-tech industrial chain resilience.

In the context of intense global technological competition, innovation capability is a core indicator of competitiveness. Therefore, from a more micro perspective, optimizing innovation resource inputs and increasing innovation outputs, thereby improving innovation efficiency, is key to enhancing high-tech industrial chain resilience. Innovation efficiency refers to the ratio of technological innovation input resources to technological innovation output. When input is lower and output is higher, innovation efficiency is higher, and vice versa (Wang et al., 2020). Digital infrastructure construction improves innovation efficiency in two main ways: in terms of innovation input, digital infrastructure construction promotes the widespread application of new information technologies such as artificial intelligence and big data, broadening information access channels, improving information accessibility and transparency, and continuously advancing the construction of digital sharing platforms. On the one hand, high-tech enterprises can select more suitable R&D personnel from the labor market to participate in the innovation process. The precise and efficient talent matching mechanism helps improve innovation efficiency (Ma et al., 2024). At the same time, enterprises can develop online learning platforms to enhance the knowledge accumulation efficiency of R&D personnel, significantly improving their overall labor quality. On the other hand, the construction of digital sharing platforms enhances the high-tech industrial chain's perception of the external environment. This increases the accessibility of scientific research resources, and the

This is a provisional file, not the final typeset article

reduction in non-productive costs allows more funds to be invested in the actual R&D of new products and technologies, effectively promoting the accumulation, integration, and absorption of innovation resources (Ma et al., 2024). More importantly, it helps the high-tech industrial chain identify short-term market innovation opportunities and its own development shortcomings, clarifying future innovation needs and achieving optimal input combinations of innovation resources (Mishra & Agarwal, 2010), thereby deeply expanding the market size and penetration of the high-tech industrial chain. In terms of innovation output, digital infrastructure construction helps high-tech industries accurately identify market demand and product positioning, improving the predictive and coordination capabilities of the high-tech industrial chain and promoting product customization (Lee et al., 2024), thereby enhancing the market penetration and circulation matching of innovation outputs. As digital infrastructure construction expands and improves, digital innovation-sharing platforms will connect upstream and downstream enterprises in the high-tech industrial chain to form an innovation network, continuously promoting the integration and reorganization of innovation knowledge and expanding the boundaries of innovation cooperation. The communication efficiency between different innovation entities is significantly improved, greatly reducing the likelihood of "ineffective innovation" and maximizing effective innovation output (Anser et al., 2020). Based on the above analysis, digital infrastructure construction can significantly improve the innovation efficiency of high-tech industrial chain, thereby enhancing their risk resistance and path-recreation capabilities. Therefore, this paper proposes Hypothesis 3:

H3: Digital infrastructure construction effectively improves the innovation efficiency of high-tech industrial chain, thereby enhancing its resilience.

According to the dynamic capability theory, digital infrastructure construction enhances the ability of high-tech industrial enterprises to perceive changes in the industry and environment. Through technological empowerment, data integration, and platform authorization, it drives the digital transformation and upgrading of traditional production factors, fully leveraging the multiplier effect of data and technology on traditional factors, and promoting the resolution of resource mismatches in high-tech industrial chain, thereby achieving rational factor allocation (Zhang et al., 2024). First, in terms of labor factor allocation, digital infrastructure construction promotes the application of digital technologies such as artificial intelligence and automation, directly replacing some routine and low-skilled jobs in high-tech industrial chain and increasing the demand for high-quality talent, thereby optimizing the human capital structure of high-tech industrial chain. At the same time, big data analysis, online recruitment platforms, and other digital tools enhance the transparency of the labor market, accelerating the matching process between enterprises and employees and significantly reducing labor resource waste caused by frictional unemployment (Ma et al., 2024). Second, in terms of capital factor allocation, digital infrastructure construction has given rise to digital financial information platforms such as JD Baitiao and Ant Credit Pay, which not only broaden the financing channels for related enterprises but also reduce their credit costs. Through project screening and real-time monitoring, the development of digital finance effectively alleviates information asymmetry between supply and demand, making direct and indirect financing information more transparent (Gao et al., 2023) and promoting the flow of capital factors to high-efficiency areas of the industrial chain. With the deepening application of digital finance, capital factors will gradually flow to the lower end of the value chain of high-tech industrial chain as the marginal cost of the network economy decreases, alleviating the financing difficulties of small and micro enterprises. The effective allocation of capital factors significantly enhances the overall development scale and flexibility of high-tech industrial chain, enabling them to promptly formulate financial service solutions to fill funding gaps in the face of external shocks. The free flow and interconnection of factors help improve the high-tech industrial chain's ability to anticipate external uncertainties, enhancing its

dynamic adjustment and adaptability (Jiang & Li, 2024). Based on the above analysis, this paper proposes Hypothesis 4:

H4: Digital infrastructure construction effectively improves factor allocation level, thereby enhancing high-tech industrial chain resilience.

4 Methodology and Data Sources

4.1 Variable Selection

4.1.1 Digital Infrastructure Construction

This paper constructs a comprehensive evaluation index system for digital infrastructure construction from four dimensions: information infrastructure, integrated infrastructure, innovation infrastructure, and government support, selecting 16 secondary indicators (Table 1):

1) **Information Infrastructure**: As the core component of digital infrastructure, information infrastructure plays a crucial role in information dissemination and provides essential hardware infrastructure for various social production and life activities. This paper refers to Chang et al. (2024) and Osei (2024), selecting seven indicators to measure information infrastructure construction: the length of long-distance optical cables, mobile phone switch capacity, internet broadband access port density, mobile phone penetration rate, number of internet domain names, per capita telecommunications business volume, and the number of 5G base stations. Among these, internet broadband access port density is calculated as (number of internet broadband access ports)/(number of internet broadband access users).

2) **Integrated Infrastructure**: Integrated infrastructure reflects the dynamic evolution of the "generational leap" between old and new infrastructure. It not only directly improves the convenience of daily life in terms of "clothing, food, housing, and transportation" but also injects new momentum into the economic development of various industries. Therefore, this paper selects five indicators to measure integrated infrastructure construction: smart highway mileage, smart transportation infrastructure, smart energy infrastructure, smart medical infrastructure, and smart living infrastructure. Among these, "smart highway mileage" is calculated as the sum of the mileage of high-speed railways, highways, first-class roads, and second-class roads in each province. Smart transportation infrastructure, smart energy infrastructure, smart medical infrastructure, and smart living infrastructure are represented by the number of smart transportation enterprises, smart energy enterprises, smart medical enterprises, and smart living enterprises, respectively.

3) **Innovation Infrastructure**: Innovation infrastructure is a key digital infrastructure that promotes the smooth development of various innovation activities, enhances social productivity, and drives high-quality economic development. This paper refers to Yang & Wang (2024), selecting three indicators to measure innovation infrastructure: the number of national-level technology business incubators, the number of university research and development institutions, and the number of national university science parks.

4) **Government Support**: Currently, digital infrastructure construction in China, represented by 5G, artificial intelligence, and industrial internet, is still in its early stages. Such infrastructure requires large upfront investments, has long economic return cycles, and is heavily influenced by national policies. Therefore, digital infrastructure construction is still government-led at this stage. This paper considers this by manually collecting provincial government work reports from 2008 to 2021 and

This is a provisional file, not the final typeset article

identifying keywords related to digital infrastructure[1]. Using Python software for analysis and processing, the ratio of keywords related to digital infrastructure to the total vocabulary in each province's government work reports from 2008 to 2021 is calculated as a proxy indicator.

4.1.2 High-tech Industrial Chain Resilience

This paper interprets high-tech industrial chain resilience based on Martin's "resistance-recovery-adaptation-innovation" theoretical framework and refers to Wu (2023) and Wu et al. (2023) to construct a comprehensive evaluation index system for high-tech industrial chain resilience from three levels: "resistance and recovery capability", "adjustment and adaptation capability", and "innovation and transformation capability" (Table 2):

1) **Resistance and Recovery Capability**: This is the foundational capability of high-tech industrial chain resilience, referring to the ability of the industrial chain to withstand external shocks and disturbances and return to its original or a new state after being impacted. This paper selects indicators from three aspects: **(A) Industrial Chain Development Scale**: The larger the industrial chain development scale, the more extensive and diversified the enterprises and markets it covers, and the richer its resource and knowledge reserves. Even after being impacted by risks, the high-tech industrial chain will only be slightly affected and can recover quickly. Referring to Gu. & Zhang. (2023), this paper selects the number of high-tech industrial chain enterprises, total profits of the high-tech industrial chain, and main business income to describe the development scale of the high-tech industrial chain. **(B) Industrial Chain Dependence**: The stronger the industrial chain dependence, the more likely it is that problems in one segment of the chain will trigger a chain reaction, paralyzing the entire chain and affecting its overall resistance and recovery capabilities. Given the current tense international situation, the role of export trade in driving domestic economic growth is gradually weakening, and enhancing the autonomous control of industrial chain is imperative. This paper selects foreign direct investment and total import and export volume of the high-tech industrial chain to represent industrial chain dependence. **(C) Internal Structure of the Industrial Chain**: A reasonable industrial chain structure facilitates the rapid flow of information, efficient allocation of resources, and flexible scheduling of production processes, while an unreasonable internal structure can lead to excessive resource concentration and operational inefficiencies. Based on Wang et al.'s interpretation of high-tech industrial chain (Wang & Yang, 2005), this paper divides high-tech industrial chain into resource-based high-tech industrial chain and technology-based high-tech industrial chain according to inter-industry linkage factors[2]. The

[1] In this paper, the keywords related to *digital infrastructure* are "5G, ICT, digital economy, digital industrialization, digital life, digital technology, data technology, datatization, digital industry, digital infrastructure, data center, data sharing, data security, data service, data governance, data assetization, digital information, digitalization, information technology, information and communication technology, information technology, information industry, information economy, information infrastructure, information service, cloud service, cloud technology, cloud application, cloud platform, cloud computing, cloud, smart manufacturing, intelligent equipment, smart economy, intelligent factory, intellectualized, intelligent, intelligent economy, intelligent science and technology, intelligent technology, intelligent terminal, e-commerce, e-technology, e-government affairs, software, software engineering, software technology, mobile communication, mobile payment, communication infrastructure, iommunication technology, knowledge economy, big data, artificial intelligence, robotics, blockchain, Internet of Things, industrial Internet, Internet, Internet platforms, modern information networks, networking technology, computer technology, networking science and technology, online".

[2] *Resource-based high-tech industrial chain* refers to the high-tech industrial chain in which the enterprises in the chain have strong technical content, but the upstream and downstream relationships are determined by resources. For example, the Chinese medicine industrial chain includes the process of "R&D and standard-setting of traditional Chinese medicine - high-quality breeding of seed sources - processing of finished traditional Chinese medicine - extraction of drugs - industrialization of traditional Chinese medicine". Technology-based high-tech industrial chain refers to the high-tech industrial chain in which each enterprise in the chain is determined by technical factors. For example, the electronic

proportion of resource-based high-tech industrial chain and technology-based high-tech industrial chain is used to reflect the spatial structure of the industrial chain, calculated as (total output value of pharmaceutical manufacturing + information chemical manufacturing)/(total output value of high-tech industries) and (total output value of aviation, aerospace, and equipment manufacturing + electronic and communication equipment manufacturing + computer and office equipment manufacturing + medical equipment and instrument manufacturing)/(total output value of high-tech industries), respectively.

2) **Adjustment and Adaptation Capability:** This is the optimization capability of high-tech industrial chain resilience, referring to the ability of the industrial chain to promptly identify and address deficiencies and quickly adapt to new development environments after being impacted by risks. This paper selects indicators from three aspects: **(A) Industrial Chain Flexibility:** The speed at which the industrial chain adjusts its structure and finds feasible alternative solutions to cope with market demand fluctuations after being impacted. This paper selects the average number of employees and the industrial diversification index to reflect industrial chain flexibility. The industrial diversification index is calculated based on the HHI (Herfindahl-Hirschman Index) following He et al. (2022). **(B) Industrial Chain Development Efficiency**: Industrial chain development efficiency is positively correlated with its stress resistance, representing resource allocation optimization capability, collaborative efficiency, and the sustainable competitive advantage of the industrial chain. This paper selects project completion rate, labor productivity, and average profit rate to calculate industrial chain development efficiency (Zheng, 2023). **(C) Industrial Chain Turnover Capability**: Excellent industrial chain turnover capability represents flexible resource scheduling capability. It contributes to long-term strategic planning and promotes innovation and cooperation among different participants to enhance the operational efficiency of enterprises on the industrial chain (Chandra & Grabis, 2009). This paper uses fixed asset investment and technology introduction and transformation costs to measure this capability.

3) **Innovation and Transformation Capability**: This is the transformative capability of high-tech industrial chain resilience, referring to the ability of the industrial chain to continuously promote new models, formats, and modes to enhance its competitiveness in the face of changing external factors. This paper selects indicators from three aspects: **(A) Industrial Chain Transformation Potential**: A good innovation foundation is a prerequisite for various innovation activities, which is the key to creating a favorable innovation atmosphere and enhancing innovation capabilities. This paper selects the full-time equivalent of R&D (Research & Development) personnel, internal and external R&D expenditure, and the number of R&D institutions with R&D activities to describe the transformation potential of the high-tech industrial chain (Yang & Liu, 2024). **(B) Industrial Chain Transformation Capability**: The more innovation outputs the industrial chain produces, the higher the likelihood of breaking "path dependence", and the stronger the industrial chain transformation capability. Referring to Liu et al. (2024), this paper selects the number of patent applications, new product development projects, and valid invention patents. **(C) Industrial Chain Innovation Linkages**: This refers to the geographical spatial innovation linkages consisting of dynamic flow of innovation entities and factors across different regions. Following Ye & Qian (2021), this paper uses *an enhanced gravity model and social network analysis* model to calculate the innovation degree

information industrial chain includes the process of "network technology research and development - network equipment - transmission network - access network equipment - network integration". It contains information technology, communication technology, Internet technology and microelectronics technology.

This is a provisional file, not the final typeset article

centrality and innovation closeness centrality to reflect the innovation linkages of the high-tech industrial chain[3].

Finally, this paper uses the comprehensive evaluation index system method to measure the level of digital infrastructure construction and high-tech industrial chain resilience. To ensure the scientific accuracy of the measurement results, it combines the entropy method and the CRITIC method for weighting. The entropy method focuses on the degree of dispersion of indicators, while the CRITIC method emphasizes the contrast intensity and conflict of indicators. Figures 5 and 6 show the changes in digital infrastructure construction and high-tech industrial chain resilience from 2008 to 2021, respectively.

4.1.3 Control Variables

To eliminate the impact of omitted variables on the theoretical analysis results, this paper selects the following six control variables to include in the model:1) Government Intervention (GOV): Local governments generally promote the development of high-tech industrial chain through policy support, park construction, market regulation, and other measures, playing a crucial role in high-tech industrial chain resilience. This paper uses the ratio of local fiscal general budget expenditure to GDP to measure this. 2) Technology Support (TECH): Direct financial support for science and technology helps promote the agglomeration and development efficiency of high-tech industrial chain. This paper uses the ratio of local fiscal science and technology expenditure to GDP to measure this. 3) Average Education Level (EDU): High-tech industries are knowledge- and technology-intensive industries. Differences in education levels affect people's understanding of high-tech industries and their ability to acquire related skills, leading to regional differences in innovation capabilities to impact high-tech industrial chain resilience. This paper uses the logarithm of the total number of people with a college degree above 6 years old to measure this. 4) Regional Entrepreneurship Vitality (ENTER): Strong entrepreneurship vitality promotes innovation and integration among upstream, midstream, and downstream segments of the high-tech industrial chain, which is crucial for enhancing the innovation and transformation capabilities of high-tech industrial chain resilience. This paper uses the logarithm of the number of urban private enterprises and self-employed individuals to measure this. 5) Foreign Dependence (FDI): Appropriate openness to foreign investment promotes the flexibility of high-tech industrial chain, but excessive reliance on foreign investment may weaken the resistance and recovery capabilities of high-tech industrial chain in the face of external risks. This paper uses the logarithm of the total investment of foreign-invested enterprises to measure this. 6) Market Consumption Scale (MAR): A larger consumption scale represents strong regional consumption and production capabilities, attracting more industrial development resources, which is important for promoting the development scale and efficiency of high-tech industrial chain. This paper uses the ratio of total retail sales of consumer goods to GDP to measure this.

4.1.4 Mediation Variables

1) Industrial Co-Agglomeration (SIA): This paper refers to the *Statistical Classification of Productive Services (2019)* issued by the National Bureau of Statistics and selects seven productive service industries closely related to high-tech industry development for calculation: real estate, finance, education, leasing and business services, transportation, warehousing and postal services, scientific research & technical services & geological exploration, and information transmission & computer services & software. The industrial co-agglomeration index is calculated as follows (Zheng & He, 2022):

[3] See Appendix A for specific calculation process.

(A) Calculate the industrial agglomeration level (location quotient) of high-tech industries and productive service industries:

$$SIAH_{i,t} = \frac{E_{iH} / E_H}{E_i / E} \tag{4-1}$$

$$SIAP_{i,t} = \frac{E_{iP} / E_P}{E_i / E} \tag{4-2}$$

Where $SIAH_{i,t}$ and $SIAP_{i,t}$ represent the industrial agglomeration levels of high-tech industries and productive service industries, respectively. E_{iH} and E_{iP} represent the number of employees in high-tech industries and productive service industries in region i, respectively. E_H and E_P represent the total number of employees in high-tech industries and productive service industries nationwide, respectively. E_i represents the total number of employees in region i, and E represents the total number of employees nationwide. For convenience, the calculation is based on urban unit employment.

(B) Calculate the industrial co-agglomeration index (SIA) based on the improved E-G index:

$$SIA_{i,t} = 1 - \frac{\left| SIAP_{i,t} - SIAH_{i,t} \right|}{SIAP_{i,t} + SIAH_{i,t}} + (SIAP_{i,t} + SIAH_{i,t}) \tag{4-3}$$

2) Innovation Efficiency (IE)

(A) Construct a three-stage DEA model evaluation index system for high-tech industrial chain innovation efficiency (Lin et al., 2021) (Table 3).

(B) Use the three-stage DEA model to measure high-tech industrial chain innovation efficiency:

① Stage 1: Use the super-efficiency SBM model as the basic model of the three-stage DEA to measure initial innovation efficiency (Du et al., 2010).

$$MIN\,\rho^* = \frac{\frac{1}{m}\sum_{i=1}^{m}\left(\bar{x} / x_{ik}\right)}{\frac{1}{s}\sum_{s=1}^{r}\left(\bar{y} / y_{sk}\right)}, \quad s.t.\left\{\begin{array}{l} \bar{x} \geq \sum_{j=1, j\neq k}^{n} x_{ij}\lambda_j, i = 1,\ldots,m \\ \bar{y} \leq \sum_{j=1, j\neq k}^{n} y_{ij}\lambda_j, s = 1,\ldots,r \\ \bar{x} \geq x_k \\ \bar{y} \leq y_k \\ \lambda_j > 0 \\ j = 1,\ldots n \end{array}\right\} \tag{4-4}$$

This is a provisional file, not the final typeset article

where, ρ^* denotes the efficiency value of the decision unit; n denotes the total number of decision units; m, r are the number of inputs and outputs of the decision unit, respectively; $\bar{x}, \bar{y}$ are the slack variables of the inputs and outputs of the decision unit, respectively; x_{ij} and y_{ij} are the actual inputs and outputs values of the decision unit, respectively, and λ_j is the corresponding weight.

② Stage 2: Construct an SFA model to analyze the causes of efficiency loss and adjust the input variables of other decision-making units based on the most efficient decision-making unit.

$$s_{ik} = f^i\left(z_k;\beta^i\right) + \delta_{ik} + \theta_{ik}, i = 1,\ldots m; k = 1,\ldots n \tag{4-5}$$

$$\hat{x}_{ik} = x_{ik} - \left[max\left(z_k\hat{\beta}^i\right) - z_k\hat{\beta}^i\right] + \left[max\left(\hat{\delta}_{ik}\right) - \hat{\delta}_{ik}\right] \tag{4-6}$$

where, s_{ik} is the difference between the ith input variable of the kth decision unit and the most efficient target input. $f^i\left(z_k;\beta^i\right)$ is the specific effect of the environmental variable on the input variable. z_k is the environmental variable, and β^i is the estimated coefficient of the environmental factor. $\delta_{ik} + \theta_{ik}$ denotes the combined error term, δ_{ik} is the management inefficiency error term, and θ_{ik} is the random error term, which are independent and uncorrelated. $\hat{x}_{ik}$ denotes adjusted input variables.

③ Stage 3: Replace the original input variables with the adjusted input variables $\hat{x}_{ik}$ and recalculate using the super-efficiency SBM model to obtain the final innovation efficiency (IE).

3) Factor Allocation Level（FAE）：

(A) Calculate the capital mismatch index φ_{Ki} and the labor mismatch index φ_{Li} :

$$\varphi_{Ki} = \frac{1}{1+\tau_{Ki}}, \varphi_{Li} = \frac{1}{1+\tau_{Li}} \tag{4-7}$$

Where φ_{Ki} and φ_{Li} are the absolute factor price distortion coefficients, which are replaced by the relative price distortion coefficients in the actual measurement process. They are calculated as follows:

$$\hat{\varphi}_{Ki} = (\frac{K_i}{K}) / (\frac{s_i\beta_{Ki}}{\beta_K}), \hat{\varphi}_{Li} = (\frac{L_i}{L}) / (\frac{s_i\beta_{Li}}{\beta_L}) \tag{4-8}$$

where s_i is the real share of economic output of region i, (K_i/K) is the real proportion of capital used in region i to the total amount of capital. (L_i/L) is the real proportion of labor used in region i to the total amount of labor. β_{Ki} and β_{Li} denote the output elasticities of capital and labor for region i,

respectively. $\beta_K = \sum_{i=1}^{N} s_i \beta_{Ki}$ and $\beta_L = \sum_{i=1}^{N} s_i \beta_{Li}$ denote the output-weighted factor contribution values of capital and labor, respectively. Equation (4-8) reflects the degree of capital and labor factor resource mismatch of region *i*. If the ratio is greater than 1 (τ < 0), it indicates over-allocation of factors in the region; conversely, if the ratio is less than 1 (τ > 0), it indicates under-allocation of factor resources in the region. In order to avoid the inconsistency of the sign direction of the regression interference, this paper takes the absolute value of τ . The larger the absolute value is, the more serious factor mismatch is.

(B) Estimate the output elasticity of capital and labor factors in each region: β_K and β_L.

From equations (4-7) and (4-8), it can be seen that the premise of calculating the capital mismatch index τ_{Ki} and the labor mismatch index τ_{Li} is to derive the output elasticity of capital and labor factors, β_K and β_L, which is specifically measured by the *Solow residual method*. Assuming that the production function is a C-D production function with constant returns to scale, the specific form is as follows:

$$Y_{it} = AK_{it}^{\beta_{Ki}} L_{it}^{1-\beta_{Ki}} \tag{4-9}$$

Where Y_{it} is the real economic output of province *i* in period *t*, measured by provincial GDP. L_{it} is the labor factor stock, expressed as the annual average number of employed people in each province. K_{it} is the capital factor stock, expressed as the fixed capital stock in each province. The source of data is the *China Statistical Yearbook* and the statistical yearbooks of each province. In this paper, the perpetual inventory method is used to calculate it:

$$K_t = I_t / P_t + (1-\sigma_t) K_{t-1} \tag{4-10}$$

Where K_t refers to the fixed capital stock in the current period (this paper takes 2008 as the base period), I_t is the nominal gross fixed capital formation in the current period, P_t is the price index of investment in fixed assets. σ_t indicates the depreciation rate, taken as 9.6%, and K_{t-1} represents the fixed capital stock in the previous period.

(C) Based on the output elasticities of capital and labor factors, β_K and β_L, calculated in (B), the capital mismatch index τ_{Ki} and the labor mismatch index τ_{Li} are derived, which in turn leads to the mediating variable FAE (factor allocation level) in this paper:

$$FAE_i = \frac{1}{2}(\tau_{Ki} + \tau_{Li}) \tag{4-11}$$

4.2 Data Sources

The data used in this paper in the process of empirical analysis come from China Statistical Yearbook, China High Technology Statistical Yearbook, China Science and Technology Statistical Yearbook, and the statistical yearbooks of each province in the past years. Some of the missing values are made up by interpolation. The descriptive statistics of each variable are analyzed in Table 4.

This is a provisional file, not the final typeset article

4.3 Methods and Models

4.3.1 Benchmark Regression Model

Based on the analysis of the impact mechanism of digital infrastructure construction on high-tech industrial chain resilience in Section 3, this subsection sets the following benchmark regression model for empirically analyzing the direct impact of digital infrastructure construction on high-tech industrial chain resilience:

$$HCR_{i,t} = \alpha_0 + \beta_0 DIC_{i,t} + \varphi_0 \sum XCV_{i,t} + \mu_i + \tau_t + \varepsilon_{i,t} \quad \text{(4-12)}$$

where, $HCR_{i,t}$ represents the level of high-tech industrial chain resilience in the region, $DIC_{i,t}$ represents the status of digital infrastructure construction in the region, and $XCV_{i,t}$ represents the control variables affecting the level of high-tech industrial chain resilience. i and t represent the region and the year, respectively. α_0 stand for the intercept term. β_0 represents the degree of the impact of digital infrastructure construction on high-tech industrial chain resilience. φ_0 represents the coefficients of a series of control variables. μ_i is the city fixed effects, controlling differences between regions; τ_t is the time fixed effects, controlling differences due to changes in the macro-environment. $\varepsilon_{i,t}$ is the random interference term.

4.3.2 Mediation Effect Model

In order to further test the indirect mechanism of digital infrastructure construction on high-tech industrial chain resilience, this paper constructs a mediation effect model on the basis of the benchmark regression model to conduct a profound analysis of digital infrastructure construction on high-tech industrial chain resilience:

$$HCR_{i,t} = \alpha_0 + \beta_0 DIC_{i,t} + \varphi_0 \sum XCV_{i,t} + \mu_i + \tau_t + \varepsilon_{i,t} \quad \text{(4-13)}$$

$$M_{i,t} = \alpha_1 + \beta_1 DIC_{i,t} + \varphi_1 \sum XCV_{i,t} + \mu_i + \tau_t + \varepsilon_{i,t} \quad \text{(4-14)}$$

$$HCR_{i,t} = \alpha_2 + \beta_2 DIC_{i,t} + \xi M_{i,t} + \varphi_2 \sum XCV_{i,t} + \mu_i + \tau_t + \varepsilon_{i,t} \quad \text{(4-15)}$$

Where, $M_{i,t}$ represents the mediating variables. Based on the theoretical analysis in Section 3, this subsection will conduct an in-depth analysis from the three ways of industrial co-agglomeration effect, innovation efficiency enhancement effect and factor allocation level rationalization effect generated by the construction of digital infrastructure. α_1 in equation (5-3) represents the intercept term. β_1 is the strength of the role of different mediation variables on the resilience of the high-tech industrial chain, and φ_1 represents the coefficients of a series of control variables. In equation (5-4), the direct effect is β_2, indicating the degree of direct influence of digital infrastructure construction on the resilience of the high-tech industrial chain. $\beta_1 * \xi$ denotes the indirect effect, indicating the degree of the role of each intermediary variable in the process of digital infrastructure construction

affecting the resilience of the high-tech industrial chain. The rest of the letters have the same meaning as the benchmark regression model.

5 Empirical analyses

5.1 Benchmark Regression Results

To determine the benchmark regression model, this paper also conducts a Hausman test[4]. Table 5 shows that the Hausman test value is 52.54 and significantly rejects the null hypothesis at the 1% level, so this paper uses fixed effects for panel regression. Table 6 presents the benchmark regression results for digital infrastructure construction and high-tech industrial chain resilience. Column (1) shows the regression results without any control variables, indicating that the coefficient of digital infrastructure construction (DIC) is significantly positive at the 5% level. Columns (2)-(7) use two-way fixed effects and gradually add different numbers of control variables. Column (7) includes all control variables, showing that the coefficient of digital infrastructure construction (DIC) is 0.072 and remains significant at the 10% level, indicating that digital infrastructure construction has a significant positive impact on high-tech industrial chain resilience, with an average effect of 7.2%.

The empirical results in Column (7) confirm that digital infrastructure construction significantly reduces information acquisition and transaction costs, promotes the popularization of digital technologies, and leverages the enabling role of "data" as a production factor, accelerating the digital transformation of high-tech industrial chain and effectively enhancing their resilience. This echoes the theoretical mechanism in Section 3. Regarding the control variables, the coefficients of EDU and ENTER are positive and significant at the 10% level, while GOV, TECH, FDI, and MAR are not significant. This suggests that: 1) Higher regional education levels represent more high-quality human capital, which helps improve the development efficiency and innovation level of high-tech industrial chain, thereby enhancing their resilience. 2)More private enterprises indicate higher regional entrepreneurship vitality, which stimulates the development potential of high-tech industrial chain through expanding market size and intense competition, crucial for enhancing the innovation and transformation capabilities of high-tech industrial chain resilience. 3)Although government intervention, including tax incentives and talent introduction policies, optimizes factor allocation in high-tech industrial chain, the negative but insignificant coefficient of GOV suggests that government decisions may be short-sighted and lagging, and the "performance-oriented" mechanism may have adverse effects on high-tech industrial chain resilience (Zhang et al., 2023). 4) Foreign dependence (FDI) and market consumption scale (MAR) are not significant but have positive coefficients, indicating that foreign investment helps improve factor allocation level in high-tech industrial chain, addressing development shortcomings, and has potential benefits for moving high-tech industrial chain toward the high end of the value chain. Market size expansion stimulates supply and demand, extending industrial chain nodes and potentially enhancing the risk resistance capability of high-tech industrial chain.

5.2 Robustness Tests

[4] Before conducting the benchmark regression, this paper performs a series of tests to ensure data stability. The correlation analysis results show that all variables have linear correlations but not causal relationships. The VIF (Variance Inflation Factor) values are all less than 10, indicating no multicollinearity issues. The unit root test results show that all variables are first-order stationary, indicating that the time series is stable. The detailed data will be available by contacting the authors directly.

This is a provisional file, not the final typeset article

5.2.1 Reducing Sample Size

To improve the accuracy of the research results, this paper also uses the method of reducing the sample size for re-regression: excluding the top and bottom 5% of provinces, the top and bottom 1% of provinces, and the top three provinces in digital infrastructure construction (Guangdong, Jiangsu, and Beijing). Columns (1)-(3) in Table 7 show that the coefficients of HCR (High-tech Industrial Chain Resilience) are significantly positive at the 5%, 1%, and 10% levels, respectively, preliminarily verifying the robustness of the core conclusions.

5.2.2 Replacing the Dependent Variable

The dependent variable, high-tech industrial chain resilience (HCR), is calculated using the entropy-CRITIC comprehensive weighting method. Referring to Ndubuisi et al. (2021), this paper replaces the entropy-CRITIC comprehensive weighting method with the entropy-TOPSIS comprehensive weighting method to recalculate HCR and conducts re-regression. Column (4) in Table 7 shows that the coefficient of DIC slightly decreases to 0.071 and remains significant at the 10% level, proving the robustness of the benchmark regression results.

5.2.3 Replacing the Core Explanatory Variable

Compared with the comprehensive evaluation index system constructed above, the capital stock of digital infrastructure can also largely reflect the actual situation of digital infrastructure construction. Therefore, this paper refers to Wen et al. (2024) and uses the total capital stock of emerging information, modern energy, and modern transportation facilities to re-measure digital infrastructure construction Wen et al. (2024). This paper calculates the capital stock of digital infrastructure by deflating fixed investment data related to digital infrastructure with 2008 as the base year and converting it into corresponding capital stock using the "perpetual inventory method". The specific formula is as follows:

$$K_{t+1} = K_t(1-\sigma) + I_t \tag{4-16}$$

where, the subscript t denotes time; K_t denotes the digital infrastructure capital stock in year t; K_{t+1} denotes the digital infrastructure capital stock in year (t+1). I_t denotes the amount of investment in year t. σ denotes the depreciation rate, taken as 9.6%. Column (5) in Table 7 shows the regression results after replacing digital infrastructure construction with "digital infrastructure construction capital stock". The coefficient of DIC is 0.089 and significantly positive at the 5% level, verifying the accuracy of the benchmark regression results again.

5.2.4 Excluding the Impact of Traditional Infrastructure Construction

Traditional infrastructure construction is the basic support and operational driver of high-tech industrial chain, enhancing communication efficiency between upstream and downstream segments and playing an important role in improving high-tech industrial chain resilience. In the development of high-tech industrial chain, traditional infrastructure mainly includes power, water, and communication infrastructure essential for production, as well as highways, railways, and bridges involved in logistics and transportation. This paper refers to Chao et al. (2024) and selects "per capita public transport vehicle operations (logarithmic transformation)" as a proxy variable for INFRAS (traditional infrastructure construction) to include in the regression. Column (6) in Table 7 shows that the coefficient of DIC is 0.073 and significant at the 10% level, while the coefficient of INFRAS is positive but not significant. These results suggest that traditional infrastructure construction may no longer meet the economic conditions necessary for improving high-tech industrial chain resilience,

while digital infrastructure construction has become a new engine for enhancing high-tech industrial chain resilience in China, again proving the robustness of the benchmark regression results.

5.2.5 Excluding all Municipalities from the Sample

Considering the differences between provinces and municipalities in administrative level, population density, industrial distribution, and other aspects, which may introduce different analytical dimensions and challenges, especially the economic linkages between municipalities and neighboring provincial cities (e.g., Shanghai and Suzhou in Jiangsu Province, Chongqing and Chengdu in Sichuan Province), this may introduce noise variables and cause significant deviations in the empirical results. Therefore, this paper excludes the four municipalities (Beijing, Shanghai, Tianjin, and Chongqing) from the sample and ensures that all individual units in the research sample are at the "provincial" level. Column (7) in Table 7 shows that the coefficient of DIC is 0.103 and significant at the 5% level, with both the effect size and significance level significantly higher than the benchmark regression results. These results strongly prove that digital infrastructure construction has become a new engine for improving high-tech industrial chain resilience in China, again confirming the robustness of the benchmark regression results.

5.3 Endogeneity Test

This paper refers to Hu et al. (2023) and Wang et al. (2023) and selects "the first and second lags of digital infrastructure construction" and "the number of post offices per million people in 1984 multiplied by the telephone penetration rate (units per 100 people) in the previous year (logarithmic transformation)" as instrumental variables, represented by "L.DIC", "L2.DIC", and "DIC2", respectively. These instrumental variables meet the requirements of relevance and exogeneity: (1) The digital infrastructure construction index system constructed in this paper focuses on "information", "integration", and "innovation" infrastructure, emphasizing information technology. In the early stages of modern information and internet development, post offices and fixed telephones were important cornerstones of information networks. Therefore, this paper uses the number of post offices and mobile phone penetration rates from the 1980s as instrumental variables, meeting the relevance requirement. (2) With the continuous transformation of information technology and industrial structure upgrading, the number of post offices and mobile phone penetration rates from the 1980s have minimal impact on high-tech industrial chain resilience now, meeting the exogeneity requirement. The specific regression results are shown in Table 8. Columns (1)-(4) show the regression results for "L.DIC" and "L2.DIC". Columns (5)-(6) show the regression results for "DIC2". The results show that the regression coefficients and signs of digital infrastructure construction on high-tech industrial chain resilience are approximately consistent with the original estimates and remain significantly positive at the 1% level, with increased significance. This indicates that the model estimation and instrumental variable selection are reasonable and effective, and the benchmark regression results remain robust, confirming that digital infrastructure construction has a significant positive impact on high-tech industrial chain resilience.

5.4 Mediation Effect Empirical Results

5.4.1 Industrial Co-Agglomeration

Column (1) in Table 9 shows that the regression coefficient of digital infrastructure construction on industrial co-agglomeration is 1.137 and significantly positive at the 5% level, indicating that digital infrastructure construction promotes the co-agglomeration of high-tech industries and productive service industries. Digital infrastructure construction significantly reduces enterprise production and communication costs, while digital services and intelligent management gradually penetrate the

This is a provisional file, not the final typeset article

production and operation of high-tech industries, increasing the demand for jobs in productive service industries. Relying on digital platforms, it effectively promotes the collaborative cooperation and agglomeration between high-tech industries and productive service industries, further driving the deep integration and collaborative development of upstream and downstream segments of high-tech industrial chain. Column (2) shows the regression results after including both DIC and SIA in the model. Only SIA is significantly positive at the 1% level, and the coefficient of DIC decreases compared to the benchmark regression coefficient (0.039 < 0.072) but is not significant. The mediation effect is 0.03, accounting for 47.38% of the total effect. These results prove that industrial co-agglomeration plays a fully mediating role in the impact of digital infrastructure construction on high-tech industrial chain resilience. After controlling for SIA, the direct effect of digital infrastructure construction on high-tech industrial chain resilience disappears, and only the transmission path of "digital infrastructure construction→industrial co-agglomeration→improvement of high-tech industrial chain resilience" exists. In high-tech industrial chain with high knowledge and technology density and high added value, industrial co-agglomeration is particularly important due to rapid technological updates and large market demand fluctuations. Industrial co-agglomeration enhances technical exchanges and resource sharing among related enterprises, improving production efficiency and innovation capabilities, and significantly enhancing the adaptability and innovation transformation capabilities of industrial chain, thereby improving its resilience. The mediation effect of industrial co-agglomeration is established, and H2 is supported.

5.4.2 Innovation Efficiency

Column (3) in Table 9 shows that the regression coefficient of digital infrastructure construction on innovation efficiency is 0.031 and significantly positive at the 10% level. This result indicates that digital infrastructure construction not only reduces innovation input by lowering labor costs and optimizing resource input efficiency but also enhances innovation output and conversion rates by accelerating technology research and development, market application, cross-industry and cross-regional collaborative innovation, and increasing enterprise entrepreneurship vitality, thereby promoting innovation efficiency. Column (4) shows the regression results after including both DIC and IE in the model. Both are significantly positive at the 10% and 1% levels, respectively, and the coefficient of DIC slightly decreases compared to the benchmark regression coefficient (0.070 < 0.072). The mediation effect is 0.031, accounting for 3.4% of the total effect. These results prove that innovation efficiency plays a partially mediating role in the impact of digital infrastructure construction on high-tech industrial chain resilience. Although the mediation effect is relatively weak, the transmission path of "digital infrastructure construction→innovation efficiency improvement→ improvement of high-tech industrial chain resilience" still exists. The mediation effect of innovation efficiency is established, and H3 is supported.

5.4.3 Factor Allocation Level

Column (5) in Table 9 shows that the regression coefficient of digital infrastructure construction on factor allocation level is -0.435 and significantly negative at the 1% level. This indicates that digital infrastructure construction optimizes human capital allocation through internet, cloud computing, and big data technologies and promotes capital allocation by establishing more transparent, efficient, and low-cost digital financial platforms. In this process, data, as a new production factor, participates in social production process, helping enterprises quickly identify market demand, optimize production processes, and enhance innovation capabilities, significantly improving factor allocation level. Column (6) shows the regression results after including both DIC and FAE in the model. Both are significant at the 10% and 1% levels, respectively. DIC is significantly positive, and its coefficient decreases compared to the benchmark regression coefficient (0.045 < 0.072). FAE is significantly

negative, and the calculated mediation effect is 0.027, accounting for 38.06% of the total effect. These results prove that factor allocation level plays a partially mediating role in the impact of digital infrastructure construction on high-tech industrial chain resilience. The transmission path of "digital infrastructure construction → factor allocation level improvement → improvement of high-tech industrial chain resilience" exists. The mediation effect of factor allocation level is established, and H4 is supported.

6 Conclusions and Policy Implications

6.1 Conclusions

This paper uses provincial panel data from 2008 to 2021 to establish evaluation index systems for digital infrastructure construction and high-tech industrial chain resilience, and conducts empirical analysis using panel regression models and mediation effect models. The main conclusions are as follows: First, the benchmark regression coefficient of digital infrastructure construction is significantly positive at the 10% level, indicating that digital infrastructure construction promotes high-tech industrial chain resilience, with an average effect of 7.2%. The experimental conclusions remain valid after a series of endogeneity and robustness tests. Second, in addition to the direct effect, digital infrastructure construction also promotes high-tech industrial chain resilience through three mediation paths: industrial co-agglomeration, innovation efficiency, and factor allocation level. Among these, industrial co-agglomeration exhibits a fully mediating effect, while innovation efficiency and factor allocation level exhibit partially mediating effects.

6.2 Policy Implications

Based on the above conclusions, this paper extends the following policy implications:

1) Government Level: The government should enhance the enabling effect of digital infrastructure construction by optimizing policy systems, strengthening investment guidance, and promoting regional collaboration. First, the government should increase fiscal investment in core digital infrastructure such as 5G, cloud computing, artificial intelligence, and blockchain, attract social capital through special funds, tax incentives, and government subsidies. This helps optimize the financial support system, and reduce the financial pressure on enterprises during digital transformation. Second, the government should promote the co-construction and sharing of digital infrastructure in city clusters and industrial clusters, and break down regional development barriers to enhance cross-regional collaboration in high-tech industrial chain, thus improving the efficiency of factor allocation across regions. Furthermore, the government should introduce regulations and industry standards related to digital infrastructure to ensure data security, information flow, and privacy protection, build a sound digital governance system, and promote resource integration and innovation collaboration among high-tech enterprises. Finally, the government should improve talent cultivation and introduction mechanisms, promote joint talent cultivation between universities and enterprises, optimize the professional skill structure required for digital infrastructure construction and application, and support the upgrading and enhancement of high-tech industrial chain resilience.

2) Industry Level: The industry should promote collaboration among related sectors and build a digital industrial ecosystem. First, the industry should use big data, blockchain, and artificial intelligence technologies to build intelligent supply chain management platforms, optimize the collaborative efficiency of production, logistics, and warehousing in high-tech industries. This enhance the agglomeration effect of high-tech industrial chain, and reduce transaction costs and risks caused by information asymmetry. Second, the industry should deepen industry-university-research

This is a provisional file, not the final typeset article

cooperation, promote the establishment of joint innovation centers by enterprises, universities, and research institutions to improve industrial innovation efficiency, thus enhancing the overall risk resistance capability of high-tech industrial chain. Furthermore, the industry should establish a digital infrastructure sharing mechanism to reduce the cost of digital transformation for enterprises and optimize the factor allocation of high-tech industrial chain.

3) Enterprise Level: Enterprises should accelerate the digital transformation process to enhance their competitiveness and adaptability. Specifically, as the core entities of industrial chain, enterprises should promote the application of digital infrastructure construction, fully utilize industrial internet, artificial intelligence, and digital twin technologies to achieve automation, intelligence, and flexibility in production processes, and enhance their ability to adapt to external shocks. Additionally, enterprises should build big data analysis and decision support systems, optimize inventory management and resource configuration, and dynamically adjust material procurement and inventory strategies to reduce the risk of industrial chain disruption. Furthermore, enterprises should deepen cross-border cooperation, actively participate in industrial ecosystem construction, stimulate more innovation models in technology research and development and production operations, and achieve high value-added transformation. Finally, while deeply embedding digital infrastructure construction, enterprises should also focus on improving network security protection capabilities to prevent data leaks, cyber attacks, and other risks that may affect high-tech industrial chain resilience.

This paper provides an in-depth analysis of the mechanisms through which digital infrastructure construction affects high-tech industrial chain resilience, but there are still many limitations. Future research could explore the following aspects: (1) conduct empirical analysis at the city and county levels to complement macro-level research, continuously expand the factor allocation models of digital infrastructure construction, and enrich its micro-level mechanisms. (2) explore more measurement methods for high-tech industrial chain resilience to more accurately reflect the characteristics of "industrial chain". (3) The measurement results of digital infrastructure construction and high-tech industrial chain resilience show significant regional differences. Future research could further verify whether there is a nonlinear relationship between the two.

Data Availability Statement

To protect the privacy of the informants, the data for this study (anonymised interview transcripts) are not made publicly available. Arrangements for sharing the data can be made by contacting the authors directly.

Conflict of Interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Author Contributions

Shen'ao Liu: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Visualization, Writing – original draft. Xia Zhou: Writing – review & editing, Supervision, Conceptualization.

Funding

The author(s) declare that financial support was received for the research, authorship, and/or publication of this article. It was supported by funding from the Beijing Municipal Social Science Foundation [grant number: Z19112] and the Pyramid Talent Engineering Training Program Project of Beijing University of Civil Engineering and Architecture [grant number:JDJQ20200305].

Reference

Ahmad, A., Alshurideh, M., Al Kurdi, B., Aburayya, A., & Hamadneh, S. (2021). Digital transformation metrics: a conceptual view. *Journal of Management Information and Decision Sciences*, 24(7), 1-18.

Anser, M. K., Khan, M. A., Awan, U., Batool, R., Zaman, K., Imran, M., Sasmoko, Indrianti, Y., Khan, A., & Bakar, Z. A. (2020). The Role of Technological Innovation in a Dynamic Model of the Environmental Supply Chain Curve: Evidence from a Panel of 102 Countries. *Processes*, 8(9), 1033. https://doi.org/10.3390/pr8091033

Chandra, C., & Grabis, J. (2009). Configurable supply chain: framework, methodology and application. *International Journal of Manufacturing technology and management*, 17(1-2), 5-22. https://doi.org/10.1504/IJMTM.2009.023776

Chang, X., Yang, Z., & Abdullah. (2024). Digital economy, innovation factor allocation and industrial structure transformation—A case study of the Yangtze River Delta city cluster in China. *Plos one*, 19(4), e0300788. https://doi.org/10.1371/journal.pone.0300788

Chao X., Lian Y., & Yuan R. (2024). Digital infrastructure construction and industrial chain resilience: empirical analysis based on industrial chain recovery capability data. *Journal of Quantitative & Technological Economics*, 41(11), 112-131. https://doi.org/10.13653/j.cnki.jqte.20240812.001

Chen H. (2019). Influencing factors of spatial structure evolution of producer services in Beijing. *Economic Geography*, 39(4), 128-135. https://doi.org/10.15957/j.cnki.jjdl.2019.04.016

Chen, J., & Wen, H. (2023). The application of complex network theory for resilience improvement of knowledge-intensive supply chains. *Operations Management Research*, 16(3), 1140-1161. https://doi.org/10.1007/s12063-023-00365-0

Crespo, J., Suire, R., & Vicente, J. (2014). Lock-in or lock-out? How structural properties of knowledge networks affect regional resilience. *Journal of Economic Geography*, 14(1), 199-219. https://doi.org/10.1093/jeg/lbt006

Du, J., Liang, L., & Zhu, J. (2010). A slacks-based measure of super-efficiency in data envelopment analysis: A comment. *European Journal of Operational Research*, 204(3), 694-697. https://doi.org/10.1016/j.ejor.2009.12.007

Du, Y., Wang, Q., & Zhou, J. (2023). How does digital inclusive finance affect economic resilience: Evidence from 285 cities in China. *International Review of Financial Analysis*, 88, 102709. https://doi.org/10.1016/j.irfa.2023.102709

This is a provisional file, not the final typeset article

Folke, C. (2006). Resilience: The emergence of a perspective for social–ecological systems analyses. Global environmental change, 16(3), 253-267. https://doi.org/10.1016/j.gloenvcha.2006.04.002

Gao, Q., Cheng, C., & Sun, G. (2023). Big data application, factor allocation, and green innovation in Chinese manufacturing enterprises. *Technological Forecasting and Social Change*, 192, 122567. https://doi.org/10.1016/j.techfore.2023.122567

Grieser, W., Maturana, G., Spyridopoulos, I., & Truffa, S. (2022). Agglomeration, knowledge spillovers, and corporate investment. *Journal of Corporate Finance*, 77, 102289. https://doi.org/10.1016/j.jcorpfin.2022.102289

Gu C., & Zhang S. (2023). Level measurement，regional differences and convergence study of the industrial chain resilience. *Inquiry into Economic Issues*, (6), 123-139.

Guo K., Pan S., & Yan S. (2020). New Infrastructure Investment and Structural Transformation.China Industrial Economics, (03), 63-80. https://doi.org/10.19581/j.cnki.ciejournal.2020.03.014

He, D., Miao, P., & Qureshi, N. A. (2022). Can industrial diversification help strengthen regional economic resilience?. *Frontiers in Environmental Science*, 10, 987396. https://doi.org/10.3389/fenvs.2022.987396

Holling C S. (1973). Resilience and stability of ecological systems. *Annual Review of Ecology and Systematics*, 4(4): 1-23. https://doi.org/10.1146/annurev.es.04.110173.000245

Hu, J., Zhang, H., & Irfan, M. (2023). How does digital infrastructure construction affect low-carbon development? A multidimensional interpretation of evidence from China. *Journal of cleaner production*, 396, 136467. https://doi.org/10.1016/j.jclepro.2023.136467

Huang, K., Wang, K., Lee, P. K., & Yeung, A. C. (2023). The impact of industry 4.0 on supply chain capability and supply chain resilience: A dynamic resource-based view. *International Journal of Production Economics*, 262, 108913. https://doi.org/10.1016/j.ijpe.2023.108913

Jiang, C., & Jin, B. (2024). Does the development of digital economy infrastructure reduce the urban-rural income gap? Theoretical experience and empirical data from China. *Kybernetes*, 53(2), 600-619. https://doi.org/10.1108/K-12-2022-1744

Jiang, W., & Li, J. (2024). Digital transformation and its effect on resource allocation efficiency and productivity in Chinese corporations. *Technology in Society*, 78, 102638. https://doi.org/10.1016/j.techsoc.2024.102638

Kangmin, W., Yuyao, Y., Xiangyu, W., Zhengqian, L., & Hong'ou, Z. (2023). New infrastructure-lead development and green-technologies: Evidence from the Pearl River Delta, China. *Sustainable Cities and Society*, 99, 104864. https://doi.org/10.1016/j.scs.2023.104864

Le, R. (2023). Measurement and improvement path analysis of China's manufacturing industry chain resilience. *Academic Journal of Business & Management*, 5(22), 12-21. https://doi.org/10.25236/AJBM.2023.052203

Lee, V. H., Foo, P. Y., Cham, T. H., Hew, T. S., Tan, G. W. H., & Ooi, K. B. (2024). Big data analytics capability in building supply chain resilience: the moderating effect of innovation-focused complementary assets. *Industrial Management & Data Systems*, 124(3), 1203-1233. https://doi.org/10.1108/IMDS-07-2022-0411

Li M., He Y., & Pan J. (2022). Dual carbon goals, carbon tax policy, and the industrial chain resilience of China's manufacturing industry. *China population, resources and environment*, 32(9), 22-34. https://doi.org/10. 12062/cpre. 20220530

Li, Y., Xu, Y., & Wen, C. (2024). Digital pathways to sustainability: the impact of digital infrastructure in the coordinated development of environment, economy and society. *Environment, Development and Sustainability*, 1-34. https://doi.org/10.1007/s10668-024-05842-5

Liao, H., Wang, B., Li, B., & Weyman-Jones, T. (2016). ICT as a general-purpose technology: The productivity of ICT in the United States revisited. *Information Economics and Policy*, 36, 10-25. https://doi.org/10.1016/j.infoecopol.2016.05.001

Lin, S., Lin, R., Sun, J., Wang, F., & Wu, W. (2021). Dynamically evaluating technological innovation efficiency of high-tech industry in China: Provincial, regional and industrial perspective. *Socio-Economic Planning Sciences*, 74, 100939. https://doi.org/10.1016/j.seps.2020.100939

Liu, R., Ge J., & Tong X. (2020). Social Order and Modernization of National Governance in the Digital Age. *China Industrial Economics*, (09), 24-41. https://doi.org/ 10.19581/j.cnki.ciejournal.2020.09.010

Liu, X., Chen, X., & Wu, Q. (2024). Exploring the systematic coupling coordination degrees of innovation activities in China's high-tech industry. *Technology Analysis & Strategic Management*, 36(11), 3308-3325. https://doi.org/10.1080/09537325.2023.2209199

Liu, Y., Wang, H., Pang, B., & Zhao, S. (2024). Capital investment, digital economy and innovation of high-tech industries. *International Review of Financial Analysis*, 96, 103761. https://doi.org/10.1016/j.irfa.2024.103761

Ma, L., Li, X., & Pan, Y. (2023). Global Industrial Chain Resilience Research: Theory and Measurement. *Systems*, 11(9), 466. https://doi.org/10.3390/systems11090466

Ma, L., Li, X., & Pan, Y. (2024). Employee allocation efficiency in the context of the digital economy: Evidence from "Broadband China" demonstration cities. *Economic Analysis and Policy*, 82, 735-752. https://doi.org/10.1016/j.eap.2024.04.010

Ma, R., Lin, Y., & Lin, B. (2023). Does digitalization support green transition in Chinese cities? Perspective from Metcalfe's Law. *Journal of Cleaner Production*, 425, 138769. https://doi.org/10.1016/j.jclepro.2023.138769

Martin, R., & Sunley, P. (2015). On the notion of regional economic resilience: conceptualization and explanation. *Journal of economic geography*, 15(1), 1-42. https://doi.org/10.1093/jeg/lbu015

Meng Q. (2023). Chain Length System and Industrial Chain Resilience: An Empirical Test Based on Multi-period DID. *Social Sciences in Xinjiang*, (01), 55-64. https://doi.org/10.20003/j.cnki.xjshkx.2023.01.007

This is a provisional file, not the final typeset article

Mishra, A. N., & Agarwal, R. (2010). Technological frames, organizational capabilities, and IT use: An empirical investigation of electronic procurement. *Information Systems Research*, 21(2), 249-270. https://doi.org/10.1287/isre.1080.0220

Ndubuisi, G., Otioma, C., & Tetteh, G. K. (2021). Digital infrastructure and employment in services: Evidence from Sub-Saharan African countries. *Telecommunications Policy*, 45(8), 102153. https://doi.org/10.1016/j.telpol.2021.102153

Nie, C., & Wan, J. (2023). How does Internet infrastructure construction affect the urban–rural income gap? Evidence from a quasi-natural experiment in China. *Progress in Development Studies*, 23(3), 317-337.

Orenstein, P. (2021). The changing landscape of supply chain networks: an empirical analysis of topological structure. *INFOR: Information Systems and Operational Research*, 59(1), 53-73. https://doi.org/10.1080/03155986.2020.1785263

Osei, D. B. (2024). Digital infrastructure and innovation in Africa: Does human capital mediates the effect?. *Telematics and Informatics*, 89, 102111. https://doi.org/10.1016/j.tele.2024.102111

Pan, Y., & Yang, M. Y. (2024). Research on the impact of digital infrastructure on the allocation efficiency of green resources in the service industry. *American Journal of Economics and Sociology*, 83(1), 223-247. https://doi.org/10.1111/ajes.12536

Queiroz, M., Tallon, P., Coltman, T., & Sharma, R. (2020). Digital infrastructure, business unit competitiveness, and firm performance growth: The moderating effects of business unit IT autonomy. *Proceedings of the 53rd Hawaii International Conference on System Sciences.*

Reggiani, A., De Graaff, T., & Nijkamp, P. (2002). Resilience: an evolutionary approach to spatial economic systems. *Networks and Spatial Economics*, 2, 211-229. https://doi.org/10.1023/A:1015377515690

Ren, F., & Tang, G. (2024). Agglomeration effects of high-tech industries: Is government intervention justified?. Economic Analysis and Policy, 83, 685-700. https://doi.org/10.1016/j.eap.2024.07.015

Schade, P., & Schuhmacher, M. C. (2022). Digital infrastructure and entrepreneurial action-formation: A multilevel study. *Journal of Business Venturing*, 37(5), 106232. https://doi.org/10.1016/j.jbusvent.2022.106232

Simmie, J., & Martin, R. (2010). The economic resilience of regions: towards an evolutionary approach. *Cambridge journal of regions, economy and society*, 3(1), 27-43. https://doi.org/10.1093/cjres/rsp029

Tan J. (2024). Research on the mechanism of digital economy to enhance the innovation efficiency of high-tech industry in the context of big data. *Applied Mathematics and Nonlinear Sciences*, 9(1), 1-14. https://doi.org/10.2478/amns.2023.2.00878

Tian, X., & Lu, H. (2023). Digital infrastructure and cross-regional collaborative innovation in enterprises. Finance Research Letters, 58, 104635. https://doi.org/10.1016/j.frl.2023.104635

Walker, B., Holling, C. S., Carpenter, S. R., & Kinzig, A. (2004). Resilience, adaptability and transformability in social-ecological systems. *Ecology and society*, 9(2). https://www.jstor.org/stable/26267673

Wan, Q., Tang, S., & Jiang, Z. (2023). Does the development of digital technology contribute to the innovation performance of China's high-tech industry?. *Technovation*, 124, 102738. https://doi.org/10.1016/j.technovation.2023.102738

Wang X., & Yang H. (2005). Structure type, function and the cultivation tactics of the high-tech industrial chain. *Science of Science and Management of S.& T.*, 26(3), 88-93.

Wang, Y., Pan, J. F., Pei, R. M., Yi, B. W., & Yang, G. L. (2020). Assessing the technological innovation efficiency of China's high-tech industries with a two-stage network DEA approach. *Socio-Economic Planning Sciences*, 71, 100810. https://doi.org/10.1016/j.seps.2020.100810

Wen, H., Liu, Y., & Zhou, F. (2024). New-type infrastructure and urban economic resilience: Evidence from China. *International Review of Economics & Finance*, 96, 103560. https://doi.org/10.1016/j.iref.2024.103560

Whitacre, B., Gallardo, R., & Strover, S. (2014). Does rural broadband impact jobs and income? Evidence from spatial and first-differenced regressions. *The Annals of Regional Science*, 53, 649-670. https://doi.org/10.1007/s00168-014-0637-x

Wu, A., Sun, Y., Zhang, H., Sun, L., Wang, X., & Li, B. (2023). Research on Resilience Evaluation of Coal Industrial Chain and Supply Chain Based on Interval Type-2F-PT-TOPSIS. *Processes*, 11(2), 566. https://doi.org/10.3390/pr11020566

Wu, W., Ji, Z., & Liang, H. (2023). The Impact of New infrastructure Construction on Optimization and Upgrading of Industrial Structure. *Buildings,* 13(10), 2580. https://doi.org/10.3390/buildings13102580

Wu, W., Wang, S., Jiang, X., & Zhou, J. (2023). Regional digital infrastructure, enterprise digital transformation and entrepreneurial orientation: Empirical evidence based on the broadband china strategy. *Information Processing & Management*, 60(5), 103419. https://doi.org/10.1016/j.ipm.2023.103419

Wu, Z. (2023). Evaluation of Provincial Economic Resilience in China Based on the TOPSIS-XGBoost-SHAP Model. *Journal of Mathematics*, 2023(1), 6652800. https://doi.org/10.1155/2023/6652800

Yang W., & Wang H. (2024). Digital infrastructure, factor allocation efficiency and modern industrial system construction. *Statistics & Decision*, 40(09), 118-123. https://doi.org/10.13546/j.cnki.tjyjc.2024.09.020

Yang, L., & Liu, Y. (2024). The impact of digital infrastructure on industrial chain resilience: evidence from China's manufacturing. *Technology Analysis & Strategic Management*, 1-17. https://doi.org/10.1080/09537325.2024.2319614

This is a provisional file, not the final typeset article

Ye, S., & Qian, Z. (2021). The economic network resilience of the Guanzhong plain city cluster, China: a network analysis from the evolutionary perspective. *Growth and Change*, 52(4), 2391-2411. https://doi.org/10.1111/grow.12530

Zeng, G., Wu, M., & Yuan, X. (2024). Digital economy and industrial agglomeration. *Economic Analysis and Policy*, 84, 475-498. https://doi.org/10.1016/j.eap.2024.09.006

Zhang A., & Sun J. (2023). Digital infrastructure, boundary affect and regional coordinate development. *Urban Problems*, (10), 77-84. https://doi.org/10.13239/j.bjsshkxy.cswt.231008

Zhang R., Chen Z., & Meng F. (2023). Government intervene and misallocation of R&D resource. Soft Science, 37(05), 17-23. https://doi.org/10.13956/j.ss.1001-8409.2023.05.03

Zhang, L., Hu, C., Guo, C., & Wang, J. (2024). Digital economy, technical change and total factor productivity: Empirical evidence from high-tech industry in China. *Telecommunications Policy*, 48(9), 102834. https://doi.org/10.1016/j.telpol.2024.102834

Zheng H., & He Y. (2022). How does industrial co-agglomeration affect high-quality economic development? Evidence from Chengdu-Chongqing Economic Circle in China. *Journal of Cleaner Production*,371, 133485. https://doi.org/10.1016/j.jclepro.2022.133485

Zheng Y. (2023). Study on the measurement and spatial and temporal differentiation of industrial chain modernization in China. *Qinghai Social Sciences*, (03), 72-79. https://doi.org/10.14154/j.cnki.qss.2023.03.020

Zhou, F., Wen, H., & Lee, C. C. (2022). Broadband infrastructure and export growth. *Telecommunications policy*, 46(5), 102347. https://doi.org/10.1016/j.telpol.2022.102347

Zhou, J., Lan, H., Zhao, C., & Wang, W. (2024). The employment effects of digital infrastructure: firm-level evidence from the 'Broadband China' strategy. *Technology Analysis & Strategic Management*, 36(10), 2647-2661. https://doi.org/10.1080/09537325.2022.2157255

Zou, H., Ji, H., Qin, H., Guo, Y., & Yang, X. (2022). Research on the strategy evolution of knowledge innovation in an Enterprise digital innovation ecosystem: kinetic and potential perspectives. *IEEE Access*, 10, 78764-78779.

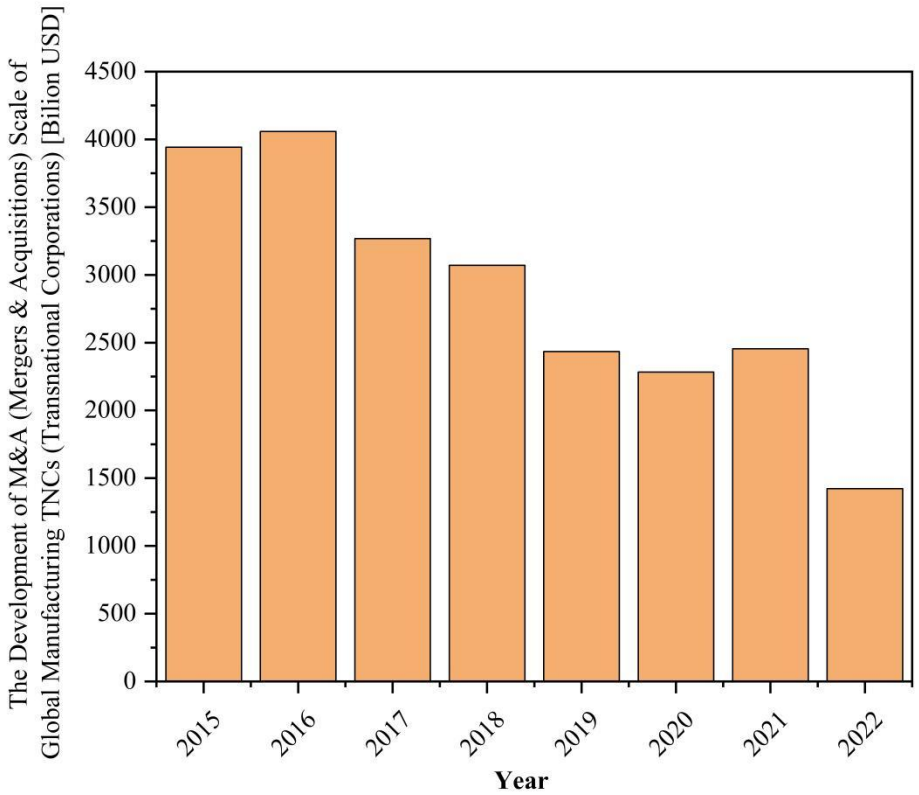

Figure 1 The Development M&A (Mergers & Acquisitions) Scale Development of Global Manufacturing TNCs (Transnational Corporations) (2015-2022)

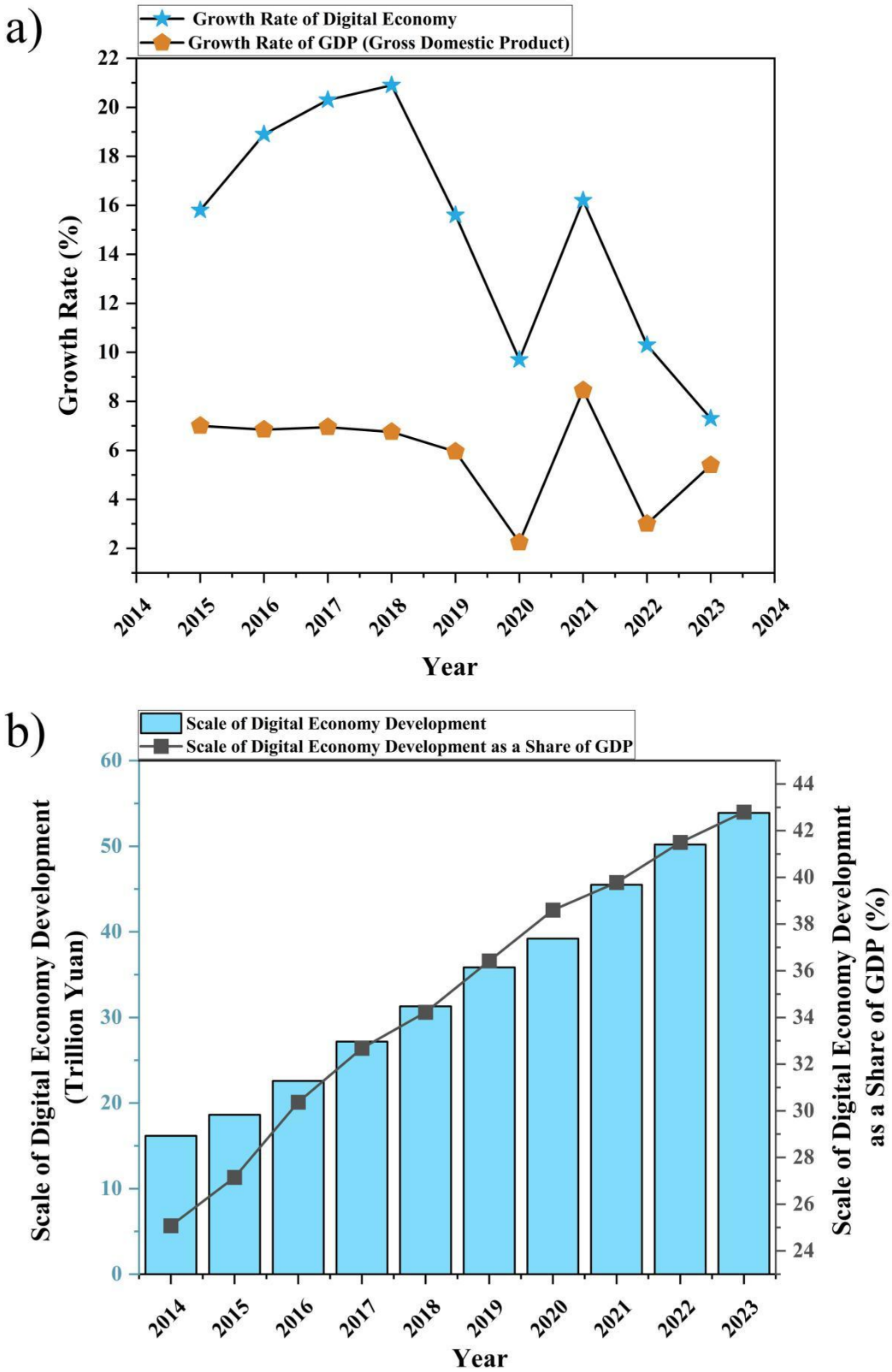

Figure 2 Development of the Digital Economy in China from 2014 to 2023

(a)Growth Rate of Digital Economy (b) Scale of Digital Economy Development

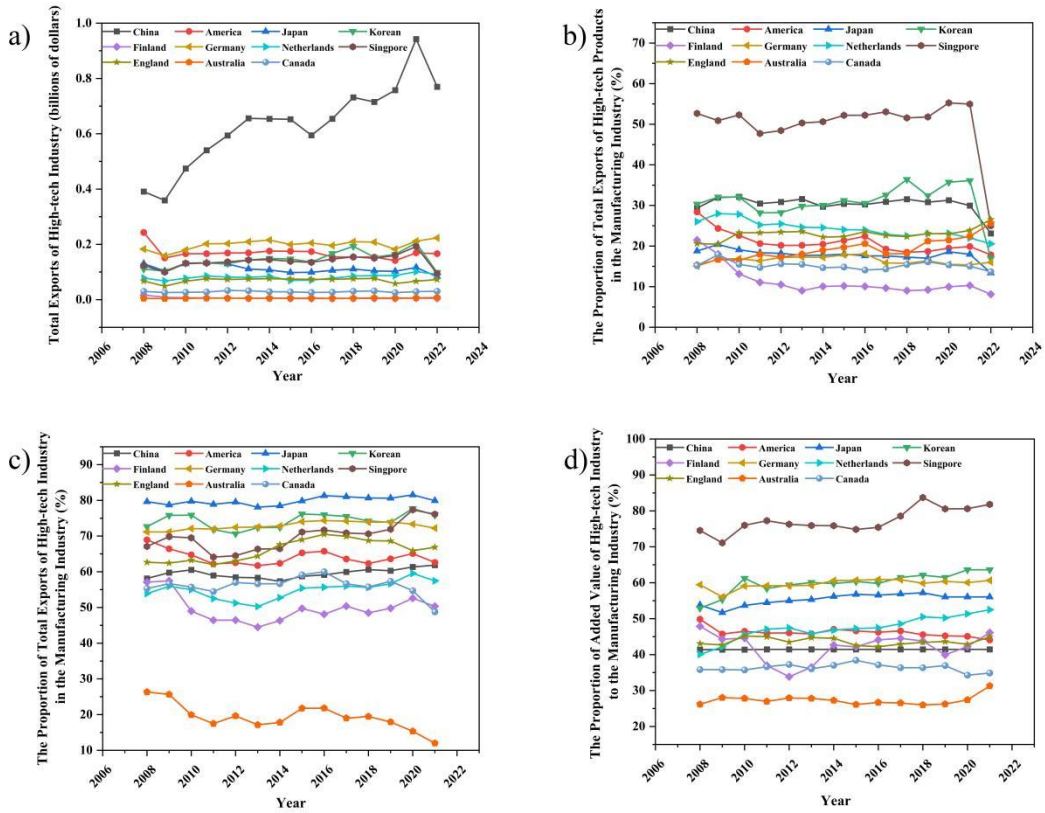

Figure 3 Development of High-tech Industry in (selected) Countries of the World

(a)Total Exports of High-tech Industry from 2008 to 2022 (b)The Proportion of Total Exports of High-tech Products in the Manufacturing Industry from 2008 to 2022 (c) The Proportion of Total Exports of High-tech Industry in the Manufacturing Industry from 2008 to 2021 (d)The Proportion of Added Value of High-tech Industry to the Manufacturing Industry from 2008 to 2021

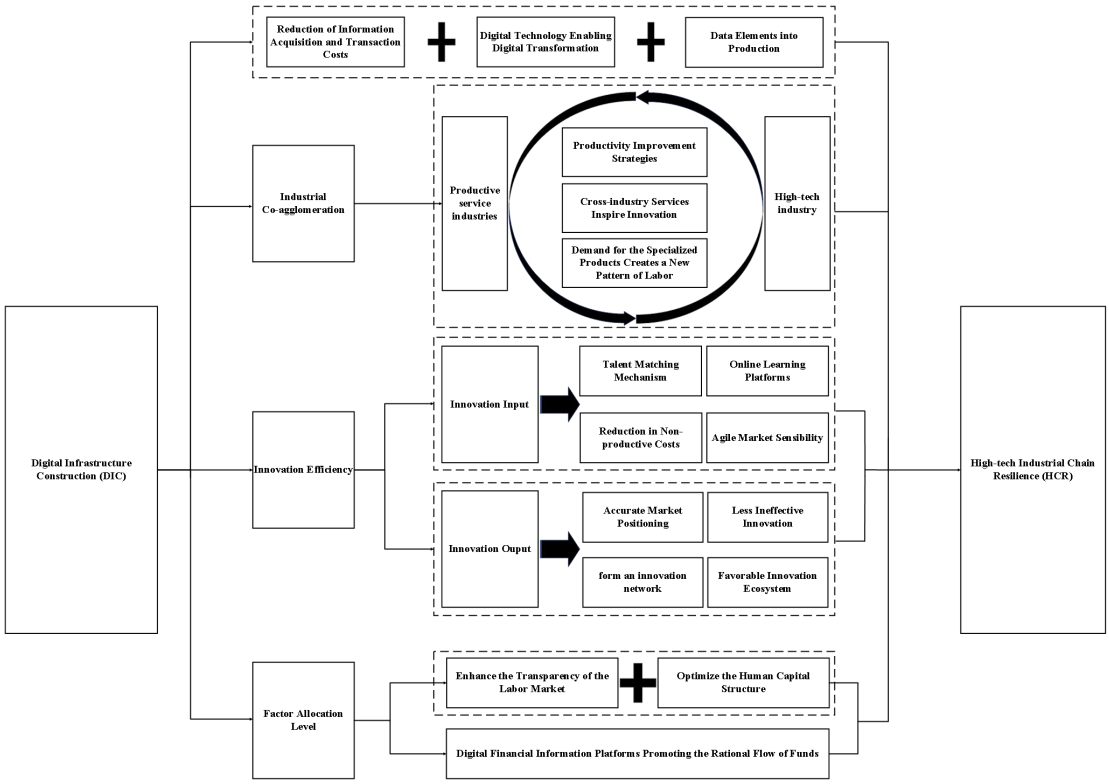

Figure 4 Map of the Mechanism of Digital Infrastructure Construction on the High-tech Industrial Chain Resilience

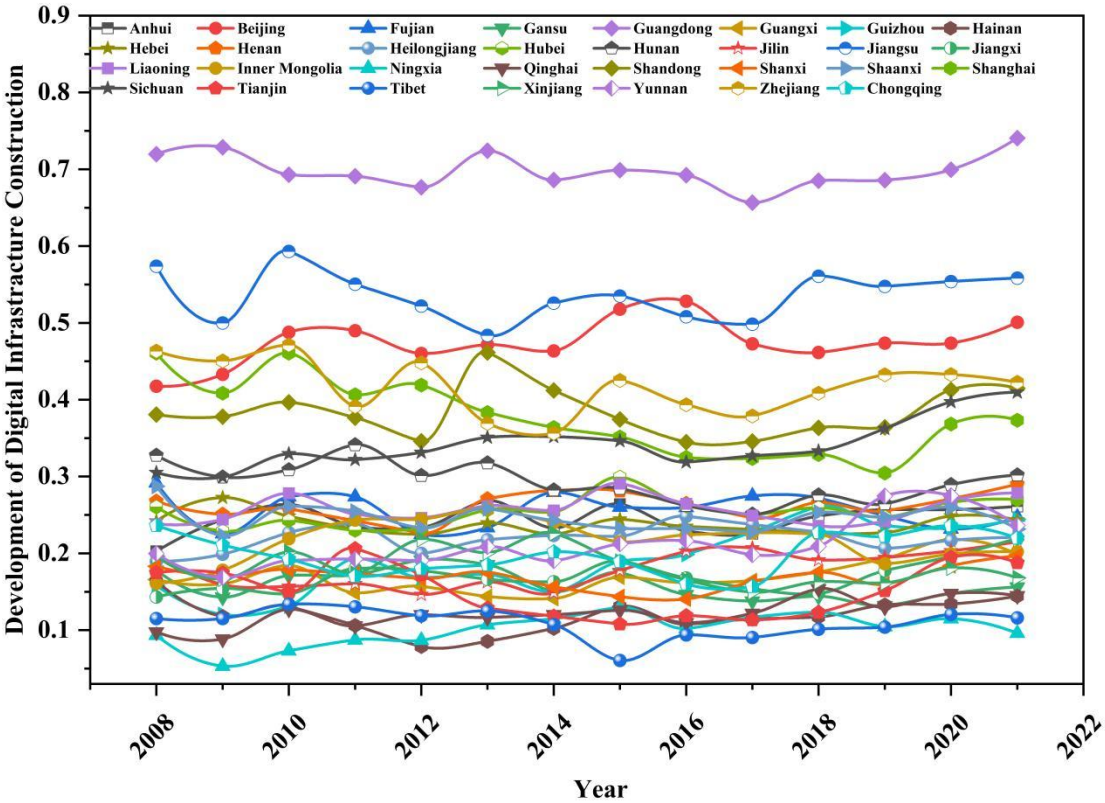

Figure 5 Comprehensive Assessment Scores of Digital Infrastructure Construction in Different Provinces

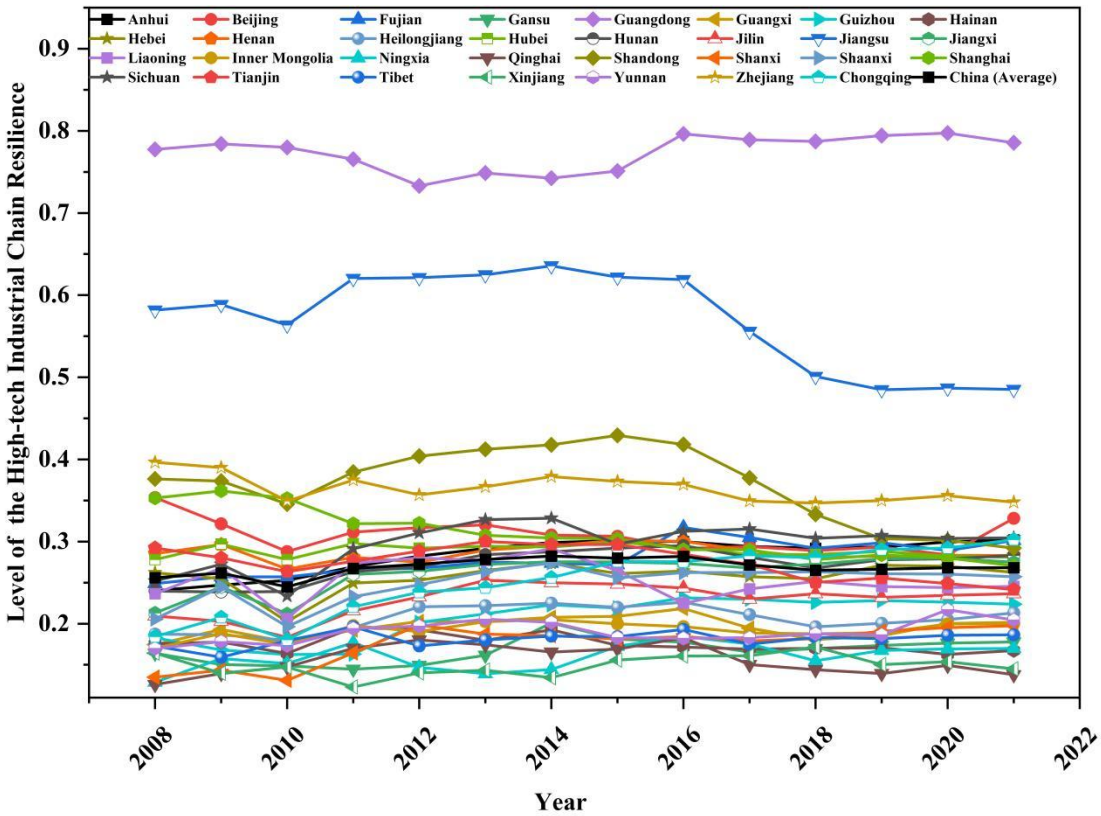

Figure 6 Comprehensive Assessment Scores of High-tech Industrial Chain Resilience in Different Provinces

Table1 Comprehensive Evaluation Indicator System of Digital Infrastructure Construction

	First Level Indicator	Second Level Indicator	Indicator Unit	Indicator Direction	Data Sources
Digital Infrastructure Construction	Information Infrastructure	the length of long-distance optical cables	ten thousand kilometers	+	*China Statistical Yearbook*、*Statistical Yearbook* of the Provinces
		,mobile phone switch capacity	ten thousand households	+	*China Statistical Yearbook*、*Statistical Yearbook* of the Provinces
		Internet broadband access port density	units/households	+	*China Statistical Yearbook*、*Statistical Yearbook* of the Provinces
		mobile phone penetration rate	/	+	*China Statistical Yearbook*、*Statistical Yearbook* of the Provinces
		the number of 5G base stations	ten thousand	+	*China Statistical Yearbook*、*Statistical Yearbook* of the Provinces
		number of internet domain names	ten thousand	+	*China Statistical Yearbook*、*Statistical Yearbook* of the Provinces
		per capita telecommunications business volume	ten thousand yuan per person	+	*China Statistical Yearbook*、*Statistical Yearbook* of the Provinces
	Integrated Infrastructure	smart highway mileage	ten thousand kilometers	+	*China Statistical Yearbook*、*Statistical Yearbook* of the Provinces
		smart transportation infrastructure	pieces	+	*Qichacha* Platform (www.qichacha.com)
		smart energy infrastructure	pieces	+	*Qichacha* Platform (www.qichacha.com)
		smart medical infrastructure	pieces	+	*Qichacha* Platform (www.qichacha.com)
		smart living infrastructure	pieces	+	*Qichacha* Platform (www.qichacha.com)
	Innovation Infrastructure	the number of national-level technology business incubators	pieces	+	*China Torch Statistical Yearbook*
		the number of university R&D (research and development) institutions	pieces	+	*China Science and Technology Statistical Yearbook*
		the number of national university science parks	pieces	+	*China Torch Statistical Yearbook*
	Government Support	the ratio of keywords related to digital infrastructure to the total vocabulary in each province's government work reports	/	+	Official website of the provincial government

Table 2 Comprehensive Evaluation Indicator System of High-tech Industrial Chain Resilience

	First Level Indicator	Second Level Indicator	Third Level Indicator	Indicator Unit	Indicator Direction	Data Sources
High-tech Industrial Chain Resilience	Resistance and Recovery Capability	Industrial Chain Development Scale	the number of high-tech industrial chain enterprises	unit	+	*China High Technology Statistical Yearbook*
			total profits of the high-tech industrial chain	trillion yuan	+	*China High Technology Statistical Yearbook*
			main business income	trillion yuan	+	*China High Technology Statistical Yearbook*
		Industrial Chain Dependence	foreign direct investment	trillion yuan	-	*China High Technology Statistical Yearbook*
			total import and export volume of the high-tech industrial chain	million dollars	+	*China High Technology Statistical Yearbook*
		Internal Structure of the Industrial Chain	the proportion of resource-based high-tech industrial chain and	/	+	*China High Technology Statistical Yearbook*
			the proportion of technology-based high-tech industrial chain	/	+	*China High Technology Statistical Yearbook*
	Adjustment and Adaptation Capability	Industrial Chain Flexibility	the industrial diversification index (HHI, Herfindahl-Hirschman Index)	/	+	*China High Technology Statistical Yearbook*
			the average number of employees	pieces	+	*China High Technology Statistical Yearbook*
		Industrial Chain Development Efficiency	project completion rate	/	+	*China High Technology Statistical Yearbook*
			labor productivity	million yuan per person	+	*China High Technology Statistical Yearbook*
			average profit rate	/	+	*China High Technology Statistical Yearbook*
		Industrial Chain Turnover Capability	fixed asset investment	trillion yuan	+	*China High Technology Statistical Yearbook*
			technology introduction and transformation costs	ten thousand yuan	+	*China Science and Technology Statistical Yearbook*
	Innovation	Industrial	the full-time	per person-year	+	*China Science and*

and Transformation Capability	Chain Transformation Potential	equivalent of R&D (Research & Development) personnel			*Technology Statistical Yearbook*
		internal and external R&D expenditure	ten thousand yuan	+	*China Science and Technology Statistical Yearbook*
		the number of R&D institutions with R&D activities	pieces	+	*China Science and Technology Statistical Yearbook*
	Industrial Chain Transformation Capability	the number of patent applications	pieces	+	*China Science and Technology Statistical Yearbook*
		new product development projects	pieces	+	*China Science and Technology Statistical Yearbook*
		valid invention patents	pieces	+	*China Science and Technology Statistical Yearbook*
	Industrial Chain Innovation Linkages	the innovation degree centrality	/	+	Calculated by *Ucinet*
		the innovation closeness centrality	/	+	Calculated by *Ucinet*

Table 3 Evaluation Index System of Three-stage DEA Model for Innovation Efficiency of High-tech Industrial Chain

	Variable Layer	**Variable Name**	**Indicator Unit**	**Data Sources**
Innovation Efficiency of the High-tech Industrial Chain	Input Layer	internal R&D expenditure	trillion yuan	*China High Technology Statistical Yearbook*
		the number of R&D personnel	person	*China High Technology Statistical Yearbook*
		Costs of technology introduction, acquisition, digestion and adaptation	trillion yuan	*China High Technology Statistical Yearbook*
		the number of R&D institutions	pieces	*China High Technology Statistical Yearbook*
	Output Layer	valid invention patents	pieces	*China High Technology Statistical Yearbook*
		Revenue from sales of new products	trillion yuan	*China High Technology Statistical Yearbook*
	Environmental Variable	Level of economic development (GDP per capita)	ten thousand yuan	*China Provincial Statistical Yearbook*
		Labor quality (number of students in general higher education)	ten thousand person	*China Provincial Statistical Yearbook*
		Degree of market competition (number of industrial enterprises above designated size)	pieces	*China Provincial Statistical Yearbook*

Table 4 Descriptive statistics of All Variables

	（1）	（2）	（3）	（4）	（5）	（6）
Variable Name	Variable Symbol	N	Mean	Sd.	Min	Max
High-tech Industrial Chain Resilience	HCR	434	0.269	0.126	0.123	0.797
Digital Infrastructure Construction	DIC	434	0.257	0.134	0.0532	0.741
Government Intervention	GOV	434	0.283	0.200	0.0998	1.354
Technology Support	TECH	434	0.005	0.003	0.002	0.014
Average Education Level	EDU	434	8.255	1.264	3.689	12.16
Regional Entrepreneurship Vitality	ENTER	434	6.366	1.014	3.384	8.712
Foreign Dependence	FDI	434	11.06	1.609	6.280	15.33
Market Consumption Scale	MAR	434	0.390	0.0630	0.220	0.610
Industrial Co-agglomeration Index	SIA	434	2.297	0.770	1.008	5.358
Innovation Efficiency	IE	434	0.218	0.273	0	1
Factor Allocation Level	FAE	434	0.329	0.234	0.0157	1.673

Table 5 Results of Hausman Test

	(1) OLS	(2) OLS	(3) FE	(4) RE
DIC	0.827***	0.794***	0.072*	0.198***
	(0.021)	(0.032)	(0.043)	(0.041)
GOV		0.065***	-0.022	-0.008
		(0.021)	(0.034)	(0.031)
TECH		-3.886***	1.262	1.441
		(1.380)	(1.019)	(1.055)
EDU		-0.005	0.021***	0.022***
		(0.003)	(0.007)	(0.007)
ENTER		0.000	0.020***	0.027***
		(0.006)	(0.007)	(0.007)
FDI		0.015***	0.002	0.004
		(0.004)	(0.003)	(0.003)
MAR		0.021	0.020	0.002
		(0.049)	(0.026)	(0.027)
CONSTANT	0.057***	0.794***	0.072*	0.198***
	(0.006)	(0.032)	(0.043)	(0.041)
N	434	434	434	434
R^2	0.780	0.792	0.260	0.792
Yearfix	NO	NO	YES	YES
Cityfix	NO	NO	YES	YES

Note: ***, * denote significant at the 1% and 10% levels, respectively; standard errors of clustering are in parentheses.

Table 6 Benchmark Regression Results on the Impact of Digital Infrastructure Construction on High-tech Industrial Chain Resilience

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
DIC	0.091**	0.089**	0.081*	0.080*	0.076**	0.073*	0.072*
	(0.044)	(0.044)	(0.044)	(0.043)	(0.043)	(0.043)	(0.043)
GOV		-0.020	-0.022	-0.025	-0.024	-0.025	-0.022
		(0.035)	(0.035)	(0.034)	(0.034)	(0.034)	(0.034)
TECH			2.075**	2.000**	1.575	1.500	1.262
			(0.981)	(0.967)	(0.964)	(0.972)	(1.019)
EDU				0.025***	0.022***	0.022***	0.021***
				(0.007)	(0.007)	(0.007)	(0.007)
ENTER					0.022***	0.021***	0.020***
					(0.007)	(0.007)	(0.007)
FDI						0.002	0.002
						(0.003)	(0.003)
MAR							0.020
							(0.026)
CONSTANT	0.246***	0.252***	0.245***	0.042	-0.071	-0.084	-0.075
	(0.011)	(0.016)	(0.016)	(0.061)	(0.069)	(0.072)	(0.074)
N	434	434	434	434	434	434	434
R^2	0.974	0.974	0.974	0.975	0.976	0.976	0.976
Yearfix	YES	YES	YES	YES	YES	YES	YES
Cityfix	YES	YES	YES	YES	YES	YES	YES

Note: ***, **,* denote significant at the 1% , 5% and 10% levels, respectively; standard errors of clustering are in parentheses.

Table 7 Results of Robustness Tests on the Impact of Digital Infrastructure Construction on High-tech Industrial Chain Resilience

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
DIC	0.092**	0.071*	0.136***	0.053***	0.092**	0.073*	0.103**
	(0.039)	(0.043)	(0.039)	(0.011)	(0.044)	(0.043)	(0.045)
INFRAS						0.012	
						(0.011)	
GOV	-0.038	-0.027	-0.025	0.006	0.014	-0.020	-0.013
	(0.031)	(0.034)	(0.030)	(0.043)	(0.035)	(0.034)	(0.034)
TECH	0.258	1.161	0.787	0.846	0.763	0.941	2.023*
	(0.926)	(1.014)	(0.963)	(1.063)	(1.050)	(1.059)	(1.064)
EDU	0.018***	0.021***	0.017***	0.021**	0.034***	0.020***	0.018**
	(0.007)	(0.007)	(0.006)	(0.009)	(0.007)	(0.007)	(0.007)
ENTER	0.016**	0.019***	0.011*	0.018***	0.018***	0.020***	0.010
	(0.006)	(0.007)	(0.006)	(0.007)	(0.007)	(0.007)	(0.007)
FDI	0.001	0.002	-0.002	0.002	0.003	0.002	0.005
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
MAR	0.037	0.021	0.054**	-0.008	0.001	0.019	-0.016
	(0.024)	(0.026)	(0.023)	(0.027)	(0.027)	(0.026)	(0.027)
CONSTANT	-0.023	-0.072	0.008	-0.562***	-0.317***	-0.182	-0.022
	(0.067)	(0.073)	(0.064)	(0.124)	(0.076)	(0.121)	(0.075)
N	434	434	392	434	434	434	378
R^2	0.969	0.976	0.930	0.977	0.984	0.976	0.979
Yearfix	YES	YES	YES	YES	YES	YES	YES
Cityfix	YES	YES	YES	YES	YES	YES	YES

Note: ***, **,* denote significant at the 1% , 5% and 10% levels, respectively; standard errors of clustering are in parentheses.

Table 8 Results of Heterogeneity Test on the Impact of Digital Infrastructure Construction on High-tech Industrial Chain Resilience

	(1) FIRST	(2) SECOND	(3) FIRST	(4) SECOND	(5) FIRST	(6) SECOND	(7) FIRST	(8) SECOND
L.DIC	0.4588***							
	(10.36)							
L2.DIC			0.2149***					
			(4.34)					
DIC2					0.1121***			
					(8.53)			
DIC3							0.0985***	
							(7.32)	
GOV	-0.0660*	-0.0236	-0.1083**	0.0178	-0.0540	0.0414	-0.0618	0.0176
	(-1.68)	(-0.62)	(-2.32)	(0.31)	(-1.12)	(0.80)	(-1.64)	(0.43)
TECH	1.1898	1.7092*	2.1319*	0.6662	2.2087*	0.3687	2.3584**	0.3451
	(1.09)	(1.66)	(1.65)	(0.46)	(1.86)	(0.29)	(2.07)	(0.28)
EDU	-0.0024	0.0212***	-0.0069	0.0236**	-0.0058	0.0215**	-0.0058	0.0217**
	(-0.31)	(2.97)	(-0.75)	(2.43)	(-0.60)	(2.11)	(-0.72)	(2.52)
ENTER	0.0056	0.0188***	0.0025	0.0161*	-0.0019	0.0201**	0.0009	0.0190**
	(0.77)	(2.78)	(0.30)	(1.83)	(-0.25)	(2.51)	(0.11)	(2.34)
FDI	0.0023	0.0003	0.0037	-0.0024	0.0011	-0.0021	0.0024	-0.0029
	(0.71)	(0.11)	(1.02)	(-0.60)	(0.31)	(-0.56)	(0.67)	(-0.76)
MAR	0.0037	0.0127	0.0091	0.0137	-0.0432	0.0153	-0.0306	0.0125
	(0.14)	(0.51)	(0.30)	(0.43)	(-1.44)	(0.50)	(-1.04)	(0.40)
DIC		0.2041**		0.6584***		0.6536***		0.7194***
		(2.27)		(2.76)		(5.28)		(4.94)
CONSTANT	0.0838	-0.0709	0.2043**	-0.1856	-0.5741***	-0.1770*	-0.1337	-0.1746**
	(1.10)	(-0.98)	(2.19)	(-1.64)	(-4.83)	(-1.92)	(-1.52)	(-2.02)
N	403	403	372	372	434	434	434	434
R^2		0.976		0.965		0.964		0.961
Phase I F-value	107.26***		18.85***		72.75***		53.56***	
Anderson-LM		93.918***		20.508***		69.009***		53.247***
Cragg-Donald		107.263		18.846		72.746		53.561
Wald F		[16.38]		[16.38]		[16.38]		[16.38]
Yearfix	YES	YES	YES	YES	YES	YES	YES	YES
Cityfix	YES	YES	YES	YES	YES	YES	YES	YES

Note: ***, **,* denote significant at the 1% , 5% and 10% levels, respectively; standard errors of clustering are in parentheses. '[]' denotes the critical value at the 10% level for the Stock-Yogo if identified test. The Anderson-LM statistic significantly rejects the original hypothesis that the instrumental variables are not identifiable. And the results of both the first-stage F-value statistic and the Cragg-Donald Wald F's test indicate that the estimation process does not suffer from a weak instrumental variable problem.

Table 9 Results of Mediation Effect on the Impact of Digital Infrastructure Construction on High-tech Industrial Chain Resilience

	(1)	(2)	(3)	(4)	(5)	(6)
	Industrial Co-agglomeration		Innovation Efficiency		Factor Allocation Level	
	SIA	HCR	IE	HCR	FAE	HCR
DIC	1.137**	0.039	0.031*	0.070*	-0.435***	0.045*
	(0.529)	(0.040)	(0.168)	(0.041)	(0.142)	(0.043)
SIA		0.030***				
		(0.004)				
IE				0.079***		
				(0.012)		
FAE						-0.063***
						(0.015)
GOV	-0.592	-0.005	0.031	-0.025	0.135	-0.014
	(0.419)	(0.032)	(0.133)	(0.032)	(0.112)	(0.033)
TECH	-1.086	1.295	18.791***	-0.223	-15.795***	0.265
	(12.540)	(0.950)	(3.977)	(0.999)	(3.364)	(1.026)
EDU	0.030	0.020***	-0.026	0.023***	0.002	0.021***
	(0.089)	(0.007)	(0.028)	(0.007)	(0.024)	(0.007)
ENTER	0.000	0.020***	0.022	0.018***	-0.036	0.018***
	(0.084)	(0.006)	(0.027)	(0.007)	(0.023)	(0.007)
FDI	0.009	0.001	-0.045***	0.005*	0.032***	0.004
	(0.039)	(0.003)	(0.012)	(0.003)	(0.010)	(0.003)
MAR	0.991***	-0.009	0.345***	-0.007	0.342***	0.042
	(0.321)	(0.025)	(0.102)	(0.025)	(0.086)	(0.026)
CONSTANT	1.387*	-0.098	0.538**	-0.099	0.216	-0.043
	(0.822)	(0.063)	(0.261)	(0.064)	(0.221)	(0.066)
SOBEL|Z|	5.914***		10.524***		2.467**	
N	434	434	434	434	434	434
R^2	0.075	0.358	0.335	0.330	0.158	0.292
Yearfix	YES	YES	YES	YES	YES	YES
Cityfix	YES	YES	YES	YES	YES	YES

Note: ***, **,* denote significant at the 1% , 5% and 10% levels, respectively; standard errors of clustering are in parentheses.

Supplementary Material

1 The Enhanced Gravity Model

This study employs an enhanced gravity model to construct innovation networks among provinces and quantify interregional innovation linkages in high-tech industrial chains. Considering the developmental characteristics of high-tech industrial chain and data availability, the innovation gravitational force between provinces is calculated as follows:

$$F_{ij} = K_{ij} \times \frac{\sqrt[4]{H_i M_i P_i Q_i} \bullet \sqrt[4]{H_j M_j P_j Q_j}}{D_{ij}^2}, K_{ij} = \frac{P_i}{P_i + P_j} \tag{1}$$

where, F_{ij} denotes the innovation gravitational force from province i to province j in high-tech industrial chain. K_{ij} represents the gravitational coefficient. Given that patent applications serve as the most direct proxy for innovation output in high-tech activities, we define it to reflect the asymmetric contributions of innovation outcomes.

H_i and H_j indicate the total R&D personnel (full-time equivalents) in provinces i and j. M_i and M_j indicate the R&D expenditure in provinces i and j. P_i and P_j represent the number of patent applications in high-tech industrial chain. Qi and Qj represent the number of valid invention patents in high-tech industrial chain. They are all sourced from the China Science and Technology Statistical Yearbook and China High Technology Statistical Yearbook. D_{ij} measures the geographical proximity between provinces, operationalized as the shortest highway distance between provincial capitals (All the data are manually extracted and validated via the website: https://www.amap.com/).

2 The Social Network Analysis Model

According to the enhanced gravity model, innovation gravitational forces are calculated for 31 Chinese provinces/municipalities/autonomous regions (2008-2021). To mitigate overfitting risks and enhance data robustness, the mean value of all calculated elements after removing extreme values is treated as threshold value, and the innovation gravity matrix of high-tech industrial chain in China is obtained. Finally, global and nodal network metrics are computed using Ucinet 6.0. Two centrality indices are selected to characterize provincial innovation linkage strength:

Degree Centrality: It is divided into out-degree centrality and in-degree centrality, which assesses the connectivity of a single individual in an innovation gravity network. An individual has a high degree centrality if it is directly connected to more other individuals. Its calculation formula is:

$$A_i = \sum_{j=1}^{N} \frac{X_{ij}}{(N-1)} \tag{2}$$

Proximity Centrality : It evaluates the shortest path distance between two different nodes, and characterizes the ability of this single individual to communicate and collaborate with other individuals quickly. The calculation formula is:

$$B_i = \frac{(N-1)}{\sum_{i=1}^{N} d_{ij}} \tag{3}$$

where, A_i and B_i denote the degree centrality and proximity centrality of node (province) i, respectively. N represents the total number of nodes (provinces) in the network. X defines the spatial adjacency matrix, where X_{ij}=1if a spatial linkage exists between node (province) i and j, and Xij=0 otherwise. dij measures the shortest-path distance between nodes (provinces) i and j.